

Adaptive network based fuzzy inference system (ANFIS) training approaches: a comprehensive survey

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Published online: 3 January 2018

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Abstract In the structure of ANFIS, there are two different parameter groups: premise and consequence. Training ANFIS means determination of these parameters using an optimization algorithm. In the first ANFIS model developed by Jang, a hybrid learning approach was proposed for training. In this approach, while premise parameters are determined by using gradient descent (GD), consequence parameters are found out with least squares estimation (LSE) method. Since ANFIS has been developed, it is used in modelling and identification of numerous systems and successful results have been achieved. The selection of optimization method utilized in training is very important to get effective results with ANFIS. It is seen that derivate based (GD, LSE etc.) and non-derivative based (heuristic algorithms such as GA, PSO, ABC etc.) algorithms are used in ANFIS training. Nevertheless, it has been observed that there is a trend toward heuristic based ANFIS training algorithms for better performance recently. At the same time, it seems to be proposed in derivative and heuristic based hybrid algorithms. Within the scope of this study, the heuristic and hybrid approaches utilized in ANFIS training are examined in order to guide researchers in their study. In addition, the final status in ANFIS training is evaluated and it is aimed to shed light on further studies related to ANFIS training.

Keywords ANFIS · ANFIS training approaches · Heuristic algorithms · Derivate based algorithms

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1 Introduction

Exhibiting intelligent behaviors of machines is provided using artificial intelligence (AI) techniques. In the literature, various AI techniques have been proposed such as fuzzy logic, neural networks, neuro-fuzzy or optimization algorithms like genetic algorithm (GA), particle swarm optimization (PSO), artificial bee colony (ABC) algorithm etc. to solve many different problem from different areas (Barchi 2016; Halim et al. 2016; Suman et al. 2016; Zahraee et al. 2016). Fuzzy logic developed by Zadeh (1965) is one of the most significant artificial intelligence techniques. Unlike the conventional set theory, it defends that membership only cannot be stated with 0 or 1. According to the fuzzy logic, each member can belong to a cluster in different membership degree. Various situations encountered in daily life include uncertainty. Fuzzy inference system allows to express the uncertain situations in the form of rules by using the decision-making mechanism. Therefore, it has been used in solving many problems (Adnan et al. 2015; Azadegan et al. 2011; Koukol et al. 2015; Lochan and Roy 2015; Suganthi et al. 2015).

One of another significant artificial intelligence techniques is artificial neural network. It models some functions realized by human brain. Artificial neural networks are formed by artificial neurons. For modelling a system using artificial neural networks, it should be trained using train data set. This called learning with samples. When the learning is completed, appropriate ANN model is obtained. Test data set is utilized to measure the success of the model. The difference between the real output and the predicted output is the error. When the error value is low, the more successful the ANN model becomes. However, the inability to explain the value of the weights belonging to ANN model is indicated as a disadvantage. Nevertheless, it has many advantages such as (1) learning using samples, (2) classification, association and pattern recognition abilities, (3) working with insufficient and incomplete information (4) realizing machine learning. So, it has used widely in the literature (Almási et al. 2016; Chojaczyk et al. 2015; Kasar et al. 2016; Vasilakos et al. 2016; Vijayalakshmi and Girish 2015).

Neuro-fuzzy is a hybrid artificial intelligence technique which is the combination of fuzzy logic and artificial neural networks. When the studies in the literature are reviewed, it can be seen that various neuro-fuzzy systems are suggested (Vieira et al. 2004). One of the most popular of these is the ANFIS proposed by Jang (1993). In ANFIS, learning ability and relational structure of the artificial neural networks with decision-making mechanism of the fuzzy logic are combined. ANFIS realizes learning with samples using train data set as in artificial neural networks. In this way, the most ideal ANFIS structure is obtained for solving the related problem. The obtained structure is subjected to the test process in order to see its impact on samples that it has never seen. The lower error values indicates the suitability of the ANFIS model. One of the most significant disadvantages of artificial neural network is that the weight values obtained could not be explained. This disadvantage is eliminated by the fuzzy inference system which is found in structure of ANFIS. Because of this structure, it is used in solving many real world problems (Kar et al. 2014).

Student modelling system (Ali and Ghatol 2004), medical system (Babazadeh et al. 2012), economic system (Fang 2012), image processing and feature extraction (Chakrapani and Soundararajan 2009), forecasting and predictions (Patil et al. 2011), manufacturing and system modelling (Topalov et al. 2009), electrical and electronics systems (Dastranj et al. 2011) are some of the fields in which ANFIS is utilized. Apart from these, different studies have been done. Cheng and Lee (1999) used a ANFIS model for fuzzy regression analysis. Loukas (2001) proposed a model based on ANFIS and hybrid learning algorithm to obtain sufficient quantitative structure-activity relationships (QSAR). Cai et al. (2003) trained ANFIS by

hybrid learning algorithm for battery state-of-charge (SOC) estimation. Lee and Gardner (2003) proposed a novel digital predistorter technique based on ANFIS. Guney and Sarikaya (2004a) calculated input resistances of thin and thick rectangular microstrip patch antennas by using ANFIS and HL. In addition to, Guney and Sarikaya (2004b, c) utilized same neuro-fuzzy method in different studies on microstrip antennas. Lee and Gardner (2006) presented adaptive digital predistorter (ADP) for RF power amplifier (PA) linearization based on ANFIS and hybrid learning algorithm. Entchev and Yang (2007) used ANFIS and hybrid algorithm to predict solid oxide fuel cell performance in residential microgeneration installation. Tabesh and Dini (2009) proposed a neuro-fuzzy model consisting ANFIS by hybrid learning algorithm for short-term water demand forecasting in Tehran. Yildiz et al. (2009) defined the application of ANFIS for vigilance level estimation by using wavelet-entropy feature extraction. Dalkilic and Apaydin (2009) utilized the validity criterion based on the fuzzy clustering. Kavitha and Duraiswamy (2011) presented an approach for the automatic screening of diabetic retinopathy in fundus images. Rajasekaran and Meena (2012) presented a neuro-fuzzy approach for magnetic-resonance (MR) brain image classification and tumour detection. Karami et al. (2013) utilized a hybrid learning algorithm to train ANFIS for modeling of heat transfer in an air-cooled heat exchanger equipped with classic twisted tape inserts. Dalkilic and Apaydin (2014) presented an approach for fuzzy regression model based on fuzzy inference system derive the satisfying solutions. Taghavifar and Mardani (2014) realized with ANFIS modeling of energy efficiency indices of agricultural tractor driving wheels. Bahadori et al. (2016) predicted combustion efficiency of natural gas in gas-fired systems with ANFIS.

In the light of the information given above, it can be seen that ANFIS is a popular and effective artificial intelligence technique. Achieving successful results with ANFIS is directly related to the optimization algorithms which it is trained. When ANFIS was first developed, hybrid learning algorithm employing GD/BP and LSE was used. In later years, it is seen that derivative based methods such as GD, LSE etc. and heuristic algorithms such as GA, PSO, ABC etc. have been utilized in ANFIS training to achieve better performance. So far, very limited review studies on ANFIS have been presented in the literature (Kabini 2011; Salleh and Hussain 2016; Walia et al. 2015a, b). Only, One of those is related to ANFIS training approaches (Salleh and Hussain 2016). A more comprehensive review is presented in this study. As a result, this survey was prepared

- to present the heuristic optimization algorithms used in ANFIS training,
- to introduce new hybrid learning approaches employed in ANFIS training,
- to evaluate the final status of ANFIS training,
- to shed light on further studies on ANFIS training.

The rest of the paper is organized as follows: in Sect. 2, ANFIS is introduced. In Sect. 3, ANFIS training approaches are presented. The derivative and heuristic based training approaches are given in Sects. 3.1 and 3.2. respectively. Finally, discussion and conclusions are provided in Sect. 4.

2 Adaptive network fuzzy inference system (ANFIS)

Network structure of ANFIS consists of two parts called as premise and consequence parts. Training ANFIS means determination of the parameters belonging to these parts utilizing an optimization algorithm. ANFIS uses the existing input–output data pairs during training. Then, IF-THEN fuzzy rules that these parts are connected each other are obtained (Jang 1993).

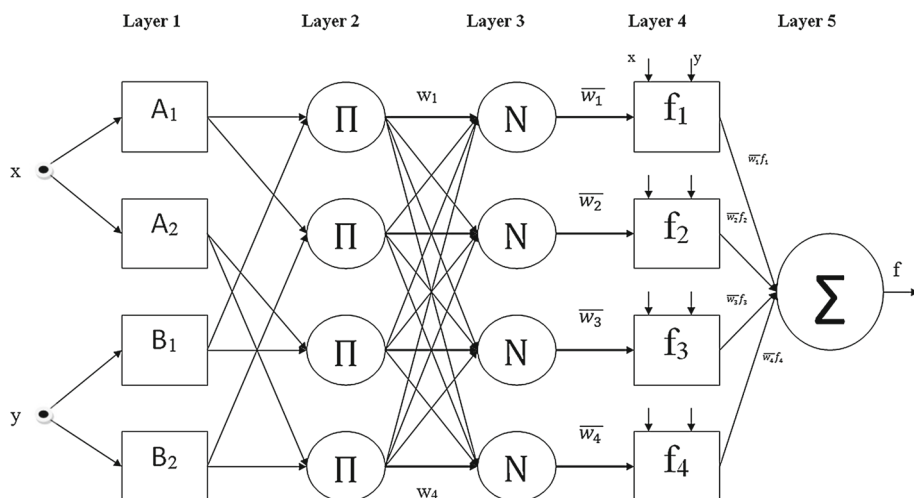


Fig. 1 The ANFIS structure that has two inputs and one output

Structure of ANFIS consists of five layers, as seen in Fig. 1. In this figure, ANFIS structure having 2 inputs and one output is given and it is composed of four membership functions and four rules. Layer structure of ANFIS is below explained according to the ANFIS structure given in Fig. 1.

2.1 Layer 1

This part is called as fuzzification layer. Fuzzification layer uses membership functions in order to obtain fuzzy clusters from input values. Parameters like determine the form of membership function and these parameters are called as premise parameter. $\{a, b, c\}$ is premise parameter set. Membership degrees of each membership function are calculated by using these parameters, as given in (1) and (2). The membership degrees obtained from this layer are shown with μ_x and μ_y .

$$\mu_{A_i}(x) = gbellmf(x; a, b, c) = \frac{1}{1 + \left| \frac{x-c}{a} \right|^{2b}} \quad (1)$$

$$O_i^1 = \mu_{A_i}(x) \quad (2)$$

2.2 Layer 2

This layer is called as rule layer. Firing strengths (w_i) for the rules are generated by using membership values computed in fuzzification layer. w_i values are found by multiplying the membership values as the following.

$$O_i^2 = w_i = \mu_{A_i}(x) \cdot \mu_{B_i}(y) \quad i = 1, 2. \quad (3)$$

2.3 Layer 3

This layer is called as normalization layer. It calculates normalized firing strengths belonging to each rule. The normalized value is the ratio of the firing strength of the i th rule to the total of all firing strengths as given in (4).

$$O_i^3 = \bar{w}_i = \frac{w_i}{w_1 + w_2 + w_3 + w_4} \quad i \in \{1, 2, 3, 4\} \quad (4)$$

2.4 Layer 4

This layer is called as defuzzification layer. Weighted values of rules are calculated in each node of this layer as given in (5). This value is determined by using first order polynomial.

$$O_i^4 = \bar{w}_i f_i = \bar{w}_i (p_i x + q_i y + r_i) \quad (5)$$

$\bar{w}_i$ is the output of the normalization layer and $\{p_i, q_i, r_i\}$ is the parameter set. These are called the consequence parameters. The number of consequence parameters of each rule is one more than the number of input.

2.5 Layer 5

It is called as the summation layer. The actual output of ANFIS is obtained by summing the outputs obtained for each rule in defuzzification layer.

$$O_i^5 = \sum_i \bar{w}_i f_i = \frac{\sum_i w_i f_i}{\sum_i w_i} \quad (6)$$

3 ANFIS training approaches

Training of ANFIS means determining the parameters in its structure using an optimization algorithm. In the training phase, premise and consequence parameters are utilized as shown in the Fig. 2. In order to obtain effective results with ANFIS, a successful training is essential. Since the first development of ANFIS, different training approaches have been suggested to achieve better performance. These approaches are divided into three as derivative based, heuristic based and hybrid.

ANFIS consists of two parts as premise and consequence. In training of these parts, we encounter two different situations according to the approaches suggested in the literature. In the first case, only one optimization algorithm is used in training of all parameters of ANFIS as shown in Fig. 3a. In the second case, hybrid learning approaches are suggested as a new method for increasing the performance. As shown in Fig. 3b, while the premise part of ANFIS are trained with an optimization algorithm, the consequence part is determined with different optimization algorithm. Thus, the problem of local minimum is eliminated and better performances is obtained. It is seen that hybrid learning approaches consisting of derivative based and heuristic algorithms are utilized widely. In addition, there are hybrid training algorithms which use only derivative based methods.

SCOPUS database is used to determine the related articles. In the database query, ANFIS and related algorithm are selected as the keywords. In addition to this, Google Scholar is also utilized to identify the methods on which there are limited number of studies.

3.1 Derivative based training approaches and applications

In the first ANFIS model presented by Jang (1993), a derivative based hybrid learning algorithm (HL) suggested as training algorithm. In HL, while premise parameters are trained with GD, consequence parameters are determined with LSE. Since the first ANFIS, HL has been used in ANFIS training mostly. When the Scope database is examined, it is seen that there

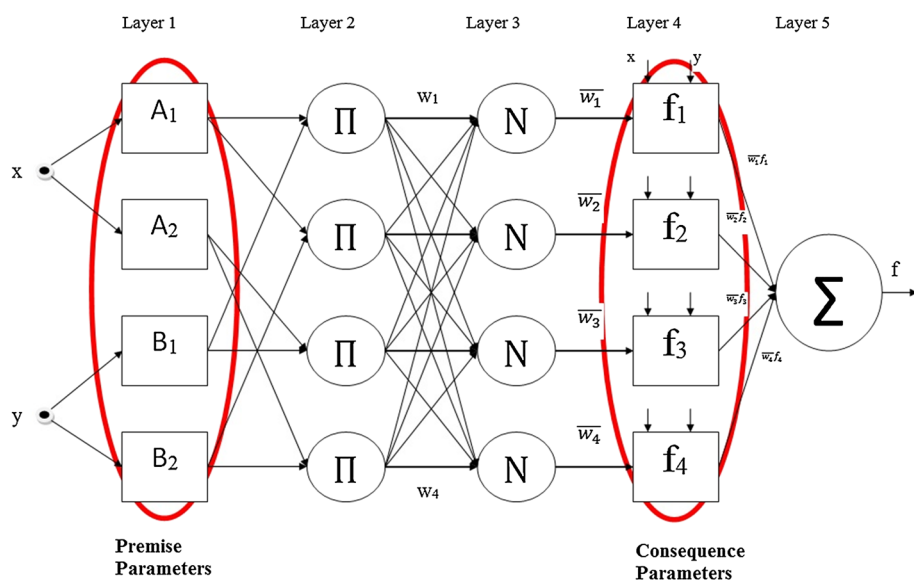


Fig. 2 Display of parameters used in ANFIS training

are thousands of ANFIS based studies. In the majority of these studies, HL is used as the training algorithm. Apart from this, it seems that a slightly different method derivative based is suggested. In this section, the approaches that all parameters are trained by derivative based methods are explained. These approaches are standard and hybrid. In this study, we focused on the different training algorithms proposed in the literature. Due to the limited different derivative based methods, this section is shortly summarized. For this purpose, the first studies on the proposed methods are given. In addition, related studies are presented in Table 1.

Apart from HL, various derivative based hybrid algorithms are also suggested, although they are few. This approaches are presented in Table 1. In some hybrid algorithms, RLSE algorithm is utilized instead of LSE (Barada and Singh 1998; Premkumar and Manikandan 2015; Xu et al. 2004). Levenberg–Marquardt (LM) algorithm based hybrid training algorithm was also applied in ANFIS training. Monirvaghefi et al. (2013) suggested a training procedure based on GD and LM for interval Type-2 ANFIS. Kalman Filter is one of the other methods used. Wang et al. (2014) proposed a multi-sensor data fusion technology based on ANFIS. Here, ANFIS was trained by using hybrid approach based on Kalman Filter (KF) and GD. In a different study, Hájek and Olej (2015) utilized a hybrid algorithm consist of KF and GD. Dehghanian Serej and Mojallali (2015) presented a ANFIS structure for speed control of elliptec motor. They determined the premise parameters of ANFIS with BP and the consequence parameters was found by extended KF (EKF). Du et al. (2015) suggested a hybrid algorithm based on square-root cubature Kalman filter (SCKF) and recursive least squares (RLS) to train ANFIS for fault prediction.

Derivative based methods used in ANFIS training are mostly hybrid, since they provide better performance. However, there are also studies in which all parameters are trained only a derivative based algorithm. Jang and Mizutani (1996) applied Levenberg–Marquardt (LM) method to train ANFIS. Liao and Zhang (2013) used LM to adjust the premise and consequence parameters of ANFIS for identification of nonlinear system. Geetha and Geethalakshmi (2012) utilized an approach based on ANFIS for detecting epileptic seizures using

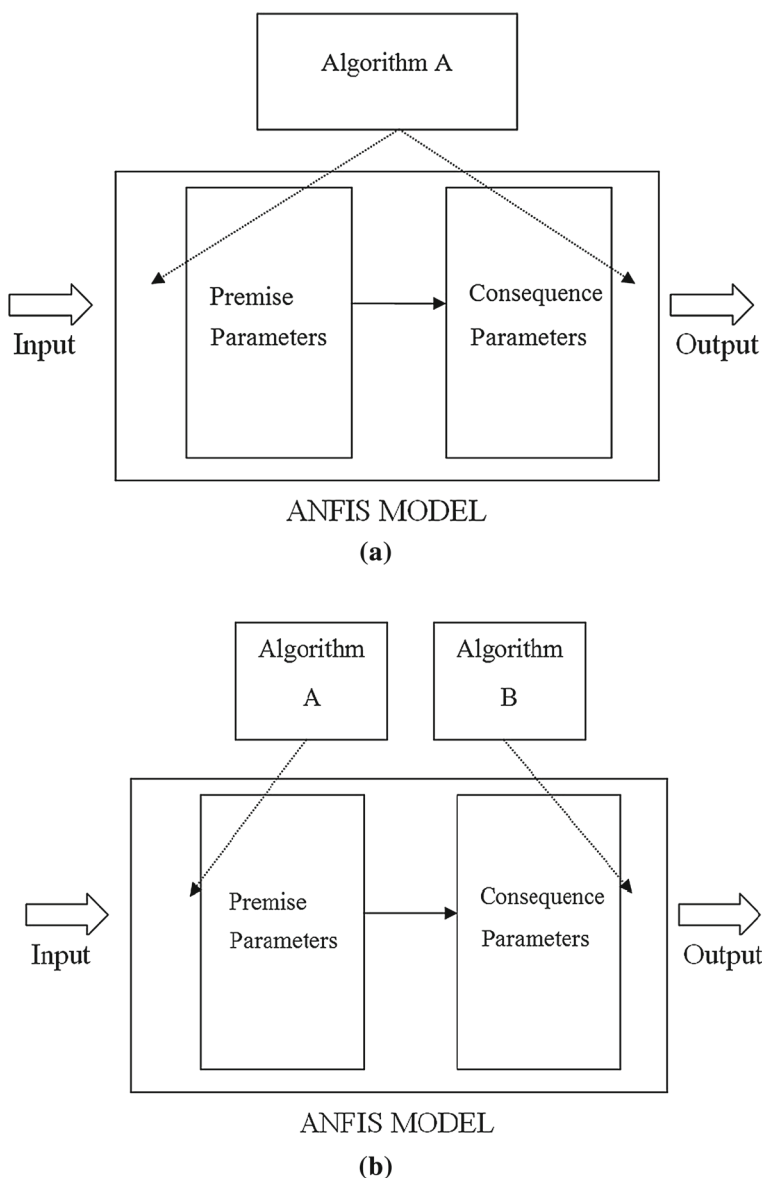


Fig. 3 Training of ANFIS by using **a** the same algorithm for both premise and consequence parts, **b** different algorithms for premise and consequence parts

electroencephalogram. Here, ANFIS was trained by modified Levenberg Marquardt (MLM) algorithm. In a different study, MLM was used to train ANFIS by Saradhadevi and Sundaram (2012). Aminifar and Yosefi (2007) adjusted the parameters of fuzzy inference system of ANFIS with BP. Chy and Uddin (2009) utilized BP algorithm to tune ANFIS parameters for the speed control of interior permanent-magnet synchronous motor (IPMSM) drives. There are different studies in which ANFIS parameters are determined with BP (Gupta and Tyagi 2014; Hassanein et al. 2013; Jhin and Hwang 2015; Tavoosi et al. 2016).

Table 1 The studies employing derivate based training of ANFIS

Year	Publication	Premise part	Consequence part	Application
1993	Jang (1993)	BP/GD	LSE	System identification
1996	Jang and Mizutani (1996)	LM	LM	System identification
1998	Barada and Singh (1998)	BP/GD	RLSE	System modelling
2004	Xu et al. (2004)	BP/GD	RLSE	A vehicle collision avoidance model
2007	Aminifar and Yosefi (2007)	BP	BP	CMOS Fuzzy Logic Controller Chip
2009	Chy and Uddin (2009)	BP	BP	Speed control of a motor
2013	Liao and Zhang (2013)	LM	LM	System identification
2013	Monirvaghefi et al. (2013)	BP/GD	LM	System identification
2014	Wang et al. (2014)	BP/GD	KF	Data fusion
2015	Premkumar and Manikandan (2015)	BP/GD	RLSE	Speed control of a motor
2015	Hájek and Olej (2015)	BP/GD	KF	The case of credit scoring
2015	Dehghanian Serej and Mojallali (2015)	BP/GD	EKF	Speed control of elliptec motor
2015	Du et al. (2015)	SCKF	RLSE	Fault prediction

In Fig. 4, different approaches used in ANFIS training are shown to better understand the chapter. As shown; BP + LSE, BP + RLSE, BP + LM, BP + KF, BP + EKF, SCKF + RLS, LM and BP are the approaches used.

3.2 Heuristic based training approaches and applications

When derivative based optimization algorithms are utilized in ANFIS training, there is a risk of local minimum. Therefore, the heuristic algorithms are used widespread in ANFIS training. When the studies in the literature are reviewed, it is seen that all parts of ANFIS or a portion of its is trained with heuristic algorithms. Standard versions and variants of heuristic algorithms are utilized for ANFIS training. The number of heuristic algorithms utilized in ANFIS training has increased rapidly in recent years. As shown in Fig. 5, the heuristic algorithms are GA, PSO, ABC, DE, HS, CS, FA, SA, MBA and AIS. Due to the fact that ANFIS training with heuristic algorithms is popular recently, this part is supported with tables and figures. All tables and figures in this survey are prepared considering the studies given. In Table 2, the algorithms utilized in training of premise and consequence parts of ANFIS are specified. In the light of this general information, ANFIS training studies based on heuristic algorithms are described below (Table 3).

3.2.1 ANFIS training studies based on GA

GA is an optimization method which models the evolution process observed in the nature. GA was first introduced in 1975 (Holland 1992). It aims to achieve the best population by using the processes such as reproduction, crossover and mutation. Since its first develop-

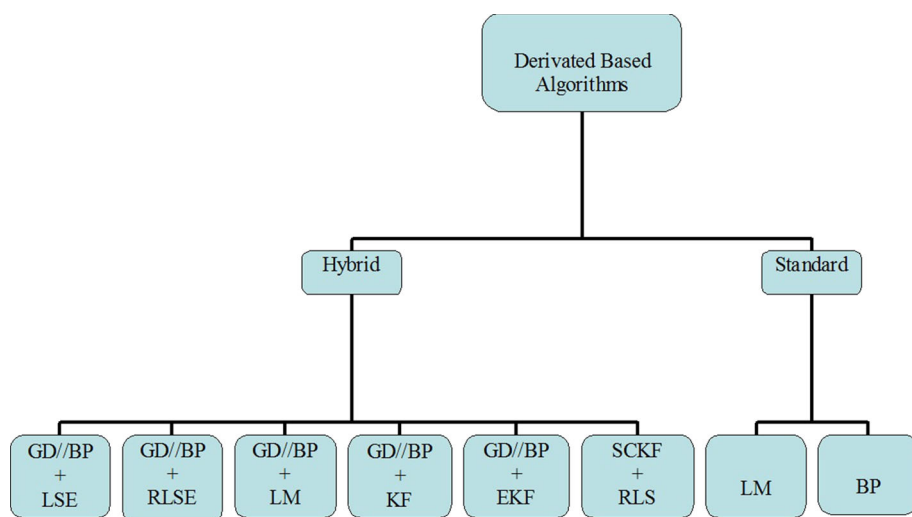


Fig. 4 ANFIS training approaches consisting of derived based algorithms

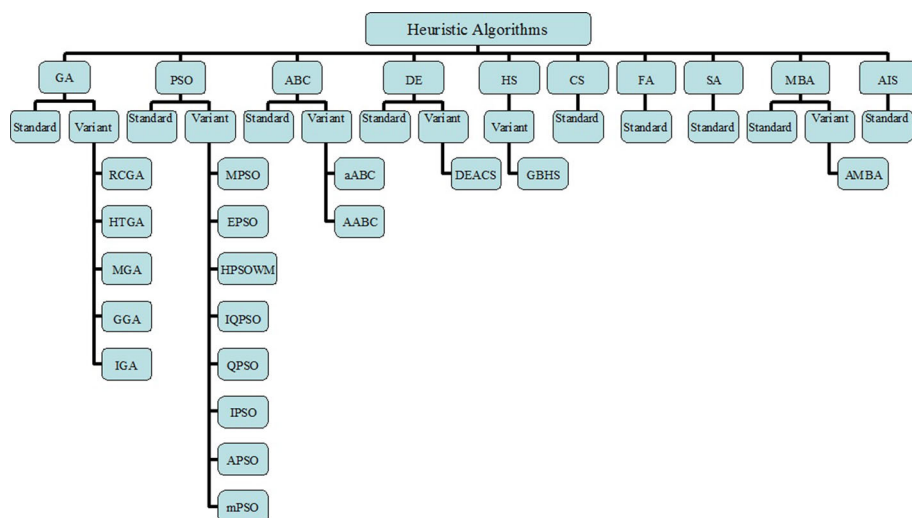


Fig. 5 ANFIS training approaches based on heuristic algorithms

ment, it has been used in solving numerous problems in various fields (Aguilar-Rivera et al. 2015; Akachukwu et al. 2014; Bhatt and Chauhan 2015; Dai et al. 2011; Karasulu and Balli 2010; Kaya and Pehlivan 2015; Sindhiya and Gunasundari 2014; Wei et al. 2014). Financial application (Aguilar-Rivera et al. 2015), power system optimization (Akachukwu et al. 2014), disease diagnosis system (Sindhiya and Gunasundari 2014), combinatorial optimisation problems (Wei et al. 2014), shop scheduling problems (Bhatt and Chauhan 2015), classification (Kaya and Pehlivan 2015), face recognition (Dai et al. 2011) and image segmentation (Karasulu and Balli 2010) are some of the areas in which GA is utilized. In addition, it is seen that it has an intensive use in ANFIS training. For ANFIS training, standard GA ve modifications

Table 2 The studies employing heuristic based training of ANFIS

Year	Publication	Premise part	Consequence part	Application
2006	Jurado et al. (2006)	GA	GA	Power quality enhancement in fuel cells
2011	Cárdenas et al. (2011)	GA	GA	Energy load profile forecast
2011	Lutfy et al. (2011b)	RCGA	RCGA	System identification
2011	Ho et al. (2011)	HTGA	HTGA	Predicting adequacy of vancomycin regimen
2011	Zhang et al. (2011)	GGA	GGA	Fault diagnosis
2011	Fang et al. (2011)	GGA	GGA	Fault diagnosis
2011	Khosravi et al. (2011)	SA	SA	Prediction
2012	Wang et al. (2012b)	DEACS	DEACS	Assessment of human operator
2012	Wang et al. (2012a)	HS	HS	Classification
2012	Turki et al. (2012)	PSO	PSO	System identification
2013	Nhu et al. (2013a)	FA	FA	Prediction of stock price
2013	Karaboga and Kaya (2013)	ABC	ABC	System identification
2014	Karaboga and Kaya (2014)	ABC	ABC	System identification
2015	Tsai et al. (2015a,b)	TBGA	TBGA	Designing
2015	Sarkheyli et al. (2015a,b)	MGA	MGA	Prediction
2015	Hussain et al. (2015)	APSO	APSO	Prediction
2016	Karaboga and Kaya (2016)	aABC	aABC	System identification
2016	Mohd and Hussain (2016)	AMBA	AMBA	Classification
2016	Suja Priyadharsini et al. (2016)	AIS	AIS	EEG

of GA is suggested. In ANFIS and GA based studies, all or part of ANFIS parameters are trained. Apart from these, there are also studies in which structural optimisation of ANFIS is made.

Hassansin et al. (2004) proposed to update the parameters of ANFIS by GA for providing automatization for INS/GPS integration systems. Samanta (2005) determined with GA the number and the parameters of membership functions used in ANFIS. Asadian et al. (2005) utilized GA to optimize the parameters of ANFIS. Oh et al. (2006) adjusted the membership parameters and the parameters of fuzzy model by GA. Qi et al. (2006) utilized GA to tune the fuzzy rules and the parameters of its for modeling and control of Direct Methanol Fuel Cell. Zanganeh et al. (2006) used GA to optimize the rule based parameters for prediction of wave parameters. Jurado et al. (2006) utilized GA to determine the premise and consequence parameters of ANFIS for power quality enhancement in fuel cells. Pan (2009) adjusted the parameter of ANFIS for forecasting classification of operating performance of enterprises. Hasan et al. (2011a) trained ANFIS by GA to predict the mislaid reading data. Cárdenas et al. (2011) determined the premise and consequence parameters of ANFIS by GA for energy load profile forecasting. Suja and Raglend (2012) optimized the membership function of fuzzy system by GA. Nasiri and Faez (2012) presented a method based on trained ANFIS by GA for extracting the Fetal Electrocardiogram (FECG) signal. Mohamed and Hameed (2015) utilized

Table 3 The studies employing derivate and heuristic based hybrid training of ANFIS

Year	Publication	Premise part	Consequence part	Application
2003	Nariman-Zadeh et al. (2003)	GA	SVD	Modelling
2004	Nariman-Zadeh et al. (2004)	GA	SVD	Modelling
2005	Mehrdad et al. (2005)	GA	SVD	Modelling
2005	Bagheri et al. (2005)	GA	SVD	Modelling
2005	Rastegar et al. (2005)	GA	LSE	System identification
2006	Shoorehdeli et al. (2006)	PSO	GD/BP	System identification
2007	Wang (2007)	PSO	LSE	–
2007	Shoorehdeli et al. (2007)	PSO	RLS	System identification
2008	Teshnehlab et al. (2008)	PSO	RLS	System identification
2009	Bagheri et al. (2009)	GA	SVD	Prediction
2009	Marzbanrad and Jamali (2009)	GA	SVD	Prediction
2009	Oliveira and Schirru (2009)	PSO	LSE	Sensor monitoring
2009	Shoorehdeli et al. (2009a)	PSO	GD/BP	System identification
2009	Shoorehdeli et al. (2009b)	PSO	EKF	System identification
2009	Shoorehdeli et al. (2009c)	PSO	FFRLS	System identification
2010	Toha and Tokhi (2010)	PSO	RLS	Modelling
2011	Jamali et al. (2011)	GA	SVD	System identification
2011	Salmalian and Soleimani (2011)	GA	SVD	Modelling of energy absorption
2011	Zangeneh et al. (2011)	DE	GD	System identification
2012	Lin et al. (2012)	QPSO	LSE	System identification
2012	Gong and Qu (2012)	MPSO	RLS	Control for motor servo system
2012	Isazadeh et al. (2012)	PSO	GD/BP	Modelling
2013	Orouskhani et al. (2013)	CSO	GD	System identification
2014	Malarvizhi et al. (2014)	PSO	EKF	System identification
2015	Wang and Ning (2015)	APAPSO	LSE	Time series prediction
2015	Mahapatra et al. (2015)	PSO	GD/BP	Motor control
2015	Mohanty and Parhi (2015)	CS	LSE	Multiple mobile robots navigation
2016	Khoshbin et al. (2016)	GA	SVD	Modelling
2016	Wanglei (2016)	IPSO	LSE	Smart home
2016	Marzi et al. (2016)	BA	LSE	System identification

a GA to adapt the parameters such as the number and shape of membership functions. Liu et al. (2013) used GA to optimize fuzzy rules and set member functions of fuzzy sets for modeling nutrient removal mechanism. Maji et al. (2013) tuned an ANFIS structure with GA for analysis and synthesis of laser forming process. Soto et al. (2013, 2015a); Soto and Melin (2014, 2015) used GA for the optimization of membership function parameters in the solution of different problems. Rai et al. (2015) optimized the parameters of ANFIS by GA for modeling of performance and emissions of a dual fuel CI engine. Vadood (2014) utilized GA to determine parameters of ANFIS for predicting the color index of acrylic fiber.

Priyadharsini and Rajan (2014b) proposed hybrid method based on ANFIS and GA for the removal of ECG artifact from the corrupted EEG signal. GA was used to tune the premise and consequence part of the ANFIS in their study. Araghi et al. (2014) utilized ANFIS and GA to set appropriate green times for traffic signal lights. In addition to the studies given here, different GA based studies were also conducted in relation to ANFIS training (Ahmadi et al. 2015; Aliabadian et al. 2015; Baghban et al. 2016; Begic Fazlic and Avdagic 2015; Begic Fazlic et al. 2015a, b, c; Dewangan et al. 2012; Haghroosta and Ismail 2015; Lu et al. 2015; Rezvani et al. 2015; Torres-Salomao et al. 2015; Wu et al. 2016; Xu et al. 2011; Yaghoobi et al. 2016; Zeinali et al. 2014; Zhang et al. 2014).

Lutfy et al. (2011b) utilized a real-coded genetic algorithm (RCGA) to optimize the premise and the consequence parameters of the ANFIS to control nonlinear multi-input multi-output systems. Apart from this, Lutfy et al. (2009, 2010a, 2011, 2011a) trained ANFIS using RCGA for solving different problems. Carrano et al. (2008) proposed a multi objective genetic algorithm (MOGA) for the training of ANFIS fuzzy networks. Ho et al. (2011) utilized a model based on ANFIS and Hybrid Taguchi-genetic algorithm (HTGA) for predicting adequacy of vancomycin regimen. They used HTGA to find the optimal premise and consequence parameters. Tsai et al. (2015a) optimized the premise and the consequence parameters of ANFIS with Taguchi-based-genetic algorithm (TBGA) for surface acoustic wave (SAW) gas sensors. In a separate study, Tsai et al. (2015b) applied the same method to optimize the micro-structure parameters of backlight modules. Zhang et al. (2011) determined the parameters of ANFIS by Gradient genetic algorithm (GGA). A separate study based on GGA was performed by Fang et al. (2011). Wang et al. (2011) proposed an ANFIS indoor positioning system based on improved GA (IGA). Rezazadeh et al. (2014) suggested a controller based on ANFIS to control a quadrotor UAV. They optimized the parameters of controller by using Nondominated Sorting Genetic Algorithm-II. Sarkheyli et al. (2015a, b) utilized a new Modified genetic algorithm (MGA) to train ANFIS.

3.2.2 ANFIS training studies based on PSO

PSO was developed with an inspiration from the swarm behaviour of birds while looking for food sources by Dr. Eberhart and Dr. Kennedy in 1995 (Kennedy and Eberhart 1995). Unlike genetic algorithm, it is a population based optimization technique. In PSO, positions of particles are used. Each particle tries to carry its position to the best position by benefiting from previous experience. PSO is one of the most popular swarm intelligence based optimization algorithms. For that reason, it is performed in solving numerous real world problems (Ashuri and Tavakolan 2011; Chao 2014; Ganguly et al. 2013; Liu and Wang 2012; Mansour et al. 2012; Ming et al. 2012; Pandey et al. 2014; Thakral et al. 2013; Yousefi et al. 2012; Zhang et al. 2015, 2010). Electrical and Electronic Engineering (Ganguly et al. 2013), Automatic Control System (Pandey et al. 2014), Communication Theory (Yousefi et al. 2012), Operations Research (Liu and Wang 2012), Mechanical Engineering (Ming et al. 2012), Civil Engineering (Ashuri and Tavakolan 2011), Fuel and Energy (Chao 2014), Medical Engineering (Zhang et al. 2010), Chemical Engineering (Thakral et al. 2013), Biological Engineering (Mansour et al. 2012) are some of the fields in which PSO is utilized. Also, PSO has an extensive use in ANFIS training. As in GA, standard PSO ve variants of PSO are utilized. With PSO based training algorithms, ANFIS is trained completely or partially. Other than this, it is seen that PSO is also applied for structure and parameter optimization of ANFIS.

Awadallah et al. (2009) used PSO to train ANFIS for current and speed controllers in a permanentmagnet brushless DC (BLDC) motor drive system. Pousinho et al. (2010) employed PSO to adjust the parameters of the membership functions of ANFIS for short-term wind

power prediction. In a separate study, they applied hybrid PSO–ANFIS approach for wind power prediction in Portugal (Pousinho and Mendes 2011). Catalão et al. (2011c) presented a hybrid approach based on wavelet transform (WT), particle swarm optimization (PSO) and ANFIS for short-term electricity prices forecasting. Zanganeh et al. (2011) utilized PSO to tune the fuzzy rule parameters of ANFIS for estimation of current-induced scour beneath marine pipelines. Sargolzaei et al. (2011) used PSO to train ANFIS for Foetal Electrocardiogram Extraction. Nasiri et al. (2011) trained ANFIS by PSO for extracting the Foetal Electrocardiogram (FECG) signal. Turki et al. (2012) determined the premise and consequence parameters of ANFIS with PSO. Rini et al. (2013) proposed training ANFIS based PSO for classification. Farid and Barakati (2014) used ANFIS training by PSO for identification of the DC motor. Chang et al. (2014) employed an ANFIS by utilizing PSO to analyse network traffic over time and tune seamless handoff properly. Priyadharsini and Rajan (2014a) used PSO to train ANFIS for eliminating ECG artifact from EEG signal. Jannaty et al. (2015) determined the parameters of ANFIS by PSO to predict scour depths. Soto et al. (2015b) utilized ANFIS and PSO for prediction of time series. Chang et al. (2015) used ANFIS and PSO to optimize the traffic flow of remote data center backup. Buyukyildiz and Tezel (2015) employed a hibrit model called PSO–ANFIS–GP for estimation of water level change of Lake Beysehir. They calculated the parameters belonging to membership functions with PSO. Ghasemi et al. (2016) proposed a hybrid approach based on ANFIS and PSO to predict peak particle velocity (PPV) due to bench blasting in open pit mines. Shamsipour et al. (2016) used ANFIS and PSO to optimize electromagnetic stirrer (EMS) process parameters. Baghban et al. (2016) utilized ANFIS to estimate true vapor pressure of volatile petroleum products. They determined the parameters belonging to the membership functions by PSO. Mustafa et al. (2016) performed ANFIS based on PSO to forecast electric load consumption. Sarica et al. (2016) suggested a new hybrid forecasting method called autoregressive adaptive network fuzzy inference system (AR–ANFIS). This model was trained by PSO. Mollaiy-Berneti (2016) proposed a model based on ANFIS by PSO for minimum miscibility pressure (MMP) estimation. Apart from these, approaches based on PSO and ANFIS was also implemented for solving different problems (Abdullah et al. 2014; Allaoua et al. 2009; Aydin et al. 2013; Azadeh and Zarrin 2016; Bakyani et al. 2016; Bassar 2015; Ch and Mathur 2010; Chang et al. 2013a,b, 2016; Çavdar 2016; Farid et al. 2013; Hasan et al. 2011b; Hosseini and Etemadi 2008; Jarraya et al. 2014; Jiang et al. 2015; Khalid et al. 2010; Rini et al. 2016; Seydi et al. 2007).

Different variations of PSO are also used in ANFIS training. Gong and Qu (2011) proposed novel algorithm based on microhabitat particle swarm optimization (MPSO) to train the parameters of ANFIS for Permanent Magnet Synchronous Motor Servo System. In a separate study, Zhang (2013) utilized MPSO to train ANFIS. Catalão et al. (2011b) suggested an intelligent system based on EPSO and ANFIS for short-term wind power forecasting in Portugal. Osório et al. (2012a) presented a hybrid approach based on wavelet transform (WT), evolutionary particle swarm optimization (EPSO) and ANFIS for short-term wind power forecasting in Portugal. EPSO and ANFIS were also used in different studies other than these (Catalão et al. 2011a; Osório et al. 2012b; Pousinho et al. 2012). Khatibinia et al. (2012) proposed an ANFIS based on fuzzy particle swarm optimization (FPSO) and fuzzy c-means (FCM). San et al. (2012) utilized hybrid particle swarm optimization with wavelet mutation (HPSOWM) for optimizing parameters belonging to ANFIS to recognize the presence of hypoglycemia. Liu et al. (2013) proposed training ANFIS with an Improved Quantum-Behaved Particle Swarm Optimization Algorithm (IQPSO). QPSO was used to optimize the network parameters of ANFIS for short-term load forecasting of microgrid by Yin (2014). Apart from these, Bagheri et al. (2014) tuned the parameter belonging to member-

ship functions of ANFIS by QPSO. Xiao et al. (2014) proposed a hybrid method combining singular spectrum analysis (SSA), ANFIS and improved particle swarm optimization (IPSO) for short-term air passenger traffic prediction. They utilized IPSO to optimize the parameters of membership functions. Hussain et al. (2015) optimized premise and consequence parameters of ANFIS by Accelerated Particle Swarm Optimization (APSO). Sharma et al. (2015) trained ANFIS by Modified Particle Swarm Optimization (mPSO) for the efficient prediction of air pollutants. Isazadeh et al. (2016) optimized the parameters of ANFIS by using multi-swarm particle swarm optimization to design wide area damping controller for the convertible static compensator (CSC).

3.2.3 ANFIS training studies based on ABC

ABC algorithm, which was developed by Karaboga in 2005, models the food searching behaviour of the honey bees (Karaboga 2005; Karaboga and Basturk 2007; Karaboga 2010). ABC algorithm is used in solving numerous real world problems in various fields and successful outcomes are obtained (Akay and Karaboga 2015; Karaboga et al. 2014; Kockanat and Karaboga 2015a, b; Koza and Karaboga 2017; Ozdemir and Karaboga 2017). Artificial neural network training (Karaboga and Kaya 2007), engineering (Karaboga 2009), image processing (Akay and Karaboga 2011), data mining (Karaboga et al. 2012), sensor networks (Okdem et al. 2011), protein structure optimization (Bahamish and Abdullah 2010) are some of the areas in which ABC algorithm is utilized. Their outside, ABC is also performed for ANFIS training. ANFIS training studies based on ABC algorithm is a more recent field of study according to GA and PSO. For that reason, the number of studies in this field is rather lower. When the literature is reviewed, it is seen that standard ABC and variants of ABC are utilized in ANFIS training.

Karaboga and Kaya (2013) proposed ANFIS training by using ABC for identification of the nonlinear static systems. Here, all parameters of ANFIS were determined with ABC algorithm. Then the results were compared with GA, BP and hybrid learning (HL) that is a combination of LSE and BP. In a separate study, the ANFIS was trained with ABC algorithm for identification of the nonlinear dynamic systems by Karaboga and Kaya (2014, 2017). The results obtained were compared with those of neuro-fuzzy based known methods frequently. In studies conducted by Karaboga and Kaya (2013, 2014, 2017), a set of premise and consequence parameters of ANFIS correspond to a food source in ABC algorithm. Pamučar et al. (2016) proposed an approach based on ANFIS by ABC for cost and risk aggregation in multi-objective route planning.

Karaboga and Kaya (2016) were developed an adaptive and hybrid artificial bee colony algorithm (aABC) to train the premise and consequence parameters of ANFIS. They used crossover rate and adaptivity coefficient in aABC algorithm, unlike the standard ABC algorithm. Better results were found with aABC according to standard ABC in ANFIS training. In different studies, Sheeja Agustin and Suresh Babu (2014, 2015) optimized the parameters of ANFIS by utilizing Adaptive Artificial Bee Colony Algorithm (AABC) algorithm.

3.2.4 ANFIS training studies based on DE

DE algorithm is a population based optimization algorithm (Storn and Price 1997). It uses similar operators as in GA. Although it has been utilized for solving numerous problems since its development (Khim Chong et al. 2014; Plagianakos et al. 2008; Xin et al. 2012), studies on ANFIS training remained limited.

Sheniha et al. (2013) proposed a novel method called ANFIS-DE (AdaptiveNeuro Fuzzy Inference System (ANFIS) tuned by differential evolution (DE) algorithm). They used DE algorithm to find the optimum parameters of ANFIS for removal of artifact from EEG signal. In their study, ANFIS-DE was outperformed with ANFIS based on general hybrid learning algorithm.

Wang et al. (2012b) developed the technique of differential evolution with ant colony search (DEACS) to optimize the parameters of ANFIS for assessment of human operator functional state. The results were compared the results obtained by using ANFIS and ANFIS-DE.

3.2.5 ANFIS training studies based on the other heuristics

GA, PSO, ABC and DE algorithms are used in ANFIS training as mentioned above. In addition to these; the different heuristic algorithm also utilized in ANFIS training, even if slightly.

Lutfy et al. (2010b) used global-best harmony search (GBHS) to train a PID-like ANFIS controller. Wang et al. (2012a) suggested a new training approach based on HS to tune all parameters of ANFIS. Araghi et al. (2016) utilized CS for design of an Optimal ANFIS Traffic Signal Controller. Mohanty et al. (2013) trained ANFIS by Firefly Algorithm. In their study, ANFIS based on FA compared with ANFIS by Hybrid Algorithm, Back Propagation and Particle Swarm Optimization (PSO). Khosravi et al. (2011) used a SA based algorithm to tune the premise and consequence parameters of the ANFIS. Hussain et al. (2016) trained ANFIS by MBA for predicting strength of Malaysian small medium enterprises. Mohd and Hussain (2016) proposed a new variant called Accelerated Mine Blast Algorithm (AMBA) to train ANFIS. Suja Priyadharsini et al. (2016) optimized the parameters of ANFIS by AIS for the elimination of the artefacts from EEG signal. Gunasekaran and Ramaswami (2011) presented a model integrating ANFIS and AIS for Forecasting Indian Stock Market. Here, AIS is used to adjust membership function parameters.

Apart from these, ANFIS was trained using derivative based and heuristic algorithms and the performances was compared. Turkmen et al. (2009) determined the design parameters of the ANFIS by four optimization algorithm: hybrid learning (HL), SA, LSE, and GA for analysis of conductor-backed asymmetric coplanar waveguides. In a separate study, Turkmen et al. (2010a) utilized these four optimization algorithms for computing the characteristic impedances of air-suspended trapezoidal and rectangular-shaped microshield lines. Training ANFIS by using these four optimization algorithms was performed by Kaya et al. (2012, 2013) and Turkmen et al. (2010b). Guney and Sarikaya (2009a) proposed an approach based on ANFIS for computing the resonant frequency of a circular microstrip antenna (MSA). They used seven optimization algorithms, LSE, nelder-mead (NM), DE, GA, HL, PSO and SA for training ANFIS and compared the results with each other. For the solution of different problems, they performed ANFIS training by different optimization algorithms (Guney and Sarikaya 2007, 2008a, b, c, 2009b). Azadeh and Zarrin (2016) utilized separately GA and PSO to train ANFIS for productivity assessment and analysis of human resource. Gocken and Boru proposed GA-ANFIS and HS-ANFIS approaches for daily mean temperature forecasting (Göçken and Boru 2016). Priyadharsini and Rajan (2014b) proposed hybrid method based on ANFIS for the removal of ECG artifact from the corrupted EEG signal. MA and GA were used to tune the premise and consequence part of the ANFIS in their study.

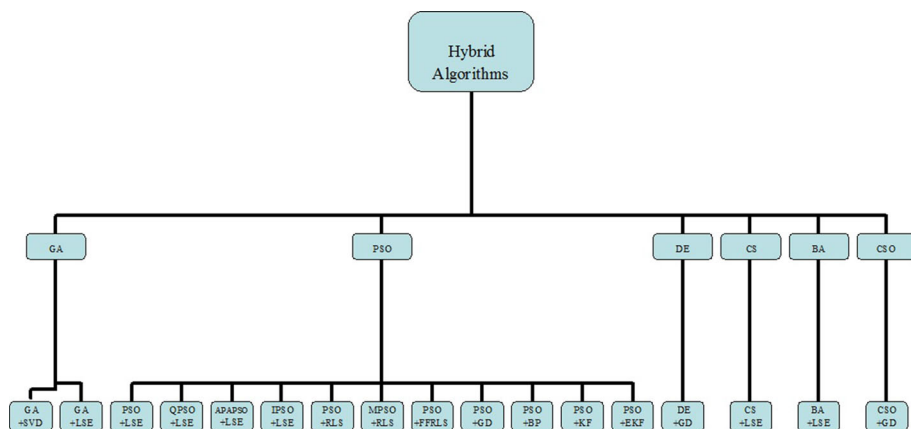


Fig. 6 Hybrid training approaches based on heuristic and derivate

3.3 Derivate and heuristic based hybrid training approaches and applications

Hybrid training algorithms have been developed by using derivative and heuristic based methods together in order to achieve more effective performances. As shown in Fig. 6, GA, PSO, DE, CS, BA, CSO are the heuristic algorithms used in hybrid training algorithms. In derivative based algorithms, LSE and GD are frequently used in hybrid training algorithms. This section are only contained heuristic and derivative based hybrid training algorithms. Hybrid training algorithms that include only derivative based algorithms are given in Sect. 3.1.

Hybrid methods are also developed to increase performance in ANFIS training by utilizing GA. In these methods, while the premise parameters of ANFIS are trained by GA, the consequence parameters are determined with a different optimization algorithm. Nariman-Zadeh et al. (2003, 2004) used a hybrid algorithm based on GA and SVD to determine the parameters of ANFIS for modeling and prediction of the explosive cutting process. Here, the parameters of Gaussian membership function in the premise part were set by GA. The consequence parameters were adjusted with SVD. Also, Mehrdad et al. (2005) used the same hybrid algorithm for modelling of the level variations of the Caspian Sea. Apart from these, GA/SVD based hybrid training approaches are applied in the different studies (Bagheri et al. 2009, 2005; Jamali et al. 2011; Khoshbin et al. 2016; Marzbanrad and Jamali 2009; Salmalian and Soleimani 2011). Rastegar et al. (2005) adjusted the parameters of membership functions by using genetic programming (GP) and GA, whereas they tuned consequent parameters with LSE.

PSO based hybrid algorithms were also utilized in ANFIS training as in GA. Wang et al. (2006) proposed a hybrid cooperative particle swarm optimization (HCP SO) for an adaptive design of ANFIS network architecture and membership functions. Wang (2007) used the hybrid algorithm which combined PSO with LSE to tune the parameters of ANFIS. Oliveira and Schirru (2009) trained ANFIS by PSO and LSE for sensor monitoring. Lin et al. (2012) proposed a novel method for training ANFIS by the quantum-behaved particle swarm optimization (QPSO) for updating the premise parameters and LSE for updating the consequence parameters. Wang and Ning (2015) suggested a hybrid method based on adaptive population activity particle swarm optimization (APAPSO) and LSE to train ANFIS. Wanglei (2016) used improved particle swarm optimization algorithm (IPSO) and LSE to obtain the premise and consequence parameters of the ANFIS in the smart home application. Apart from those,

LSE was applied in different ANFIS training studies (Jiang et al. 2013, 2012; Zuo et al. 2010). Toha and Tokhi (2010) utilized PSO to optimize the premise parameters of the ANFIS and used recursive least squares (RLS) to determine the consequence parameters of the ANFIS. Gong and Qu (2012) utilized Microhabitat particle swarm optimization (MPSO) algorithm and RLS algorithm to train ANFIS for Permanent Magnet Synchronous Motor Servo System. Shoorehdeli et al. (2007) proposed a hybrid approach based on PSO and RLS for training ANFIS. Teshnehlab et al. (2008) used the hybrid method formed fuzzy PSO with recursive least squares (RLS) for training ANFIS. Shoorehdeli et al. (2009c) presented a hybrid learning algorithm based on PSO for training the premise part and forgetting factor recursive least square (FFRLS) for training the consequence part. Isazadeh et al. (2012) utilized a hybrid approach based on PSO and GD to train ANFIS for modelling dynamic load-shedding scheme. Shoorehdeli et al. (2009a, 2006) suggested a hybrid method based on PSO for training the premise part and gradient descent (GD) for training the consequence part. Mahapatra et al. (2015) used ANFIS to control the speed of an induction motor. They utilized PSO and backpropagation algorithm in training ANFIS. Shoorehdeli et al. (2009b) proposed intelligent hybrid stable learning algorithm based on PSO and Extended Kalman Filter (EKF) for training ANFIS. They used PSO for training the premise part and utilized EKF to train the consequence part. In addition, Malarvizhi et al. (2014) suggested a hybrid learning algorithm based on PSO and EKF to determine the premise and consequence parameters of ANFIS.

Zangeneh et al. (2011) utilized a hybrid learning algorithm based on DE and GD to determine premise and consequence parameters of ANFIS. In the study, The basic DE (DE/rand/1/bin) was used for training the premise part and GD was employed for training the consequence part. Parhi (2015) applied a hybrid approach based on CS for training the premise part and LSE for consequence part of ANFIS. The approach was implemented for single and multiple mobile robots in static environments. The obtained results compared with different methods based on neuro-fuzzy and fuzzy. Marzi et al. (2016) proposed a training algorithm based on BA and LSE for ANFIS. BA was used for adjusting the premise parameters of ANFIS. In addition to LSE was utilized to tune the consequence parameters of its. In their study, they compared the ANFIS based on BA-LSE with ANFIS. Here, ANFIS based on BA-LSE enhanced local minima avoidance. Orouskhani et al. (2013) used a hybrid method based on CSO and GD for training ANFIS. The premise parameters were trained with CSO, while the consequence parameters were determined by using GD.

4 Discussion and conclusions

It is seen that derivative based methods and heuristic algorithms are used in ANFIS training. In the first ANFIS structure, a hybrid method consisting of GD and LSE was utilized. Various methods have been suggested to obtain better performances. Proposed methods are discussed three topics as derivative, heuristic and hybrid.

GD + LSE based hybrid algorithm is the first training algorithm suggested for ANFIS. GD + LSE based hybrid algorithm generally provides higher performance than other derivative based approaches. It is used extensively today. Apart from GD + LSE based hybrid algorithm known as HL, BP + LSE, BP + RLSE, BP + LM, BP + KF, BP + EKF, SCKF + RLS, LM and BP are the approaches used. The terminology list of abbreviations used in this study is given in Table 4.

Heuristic algorithms have been successfully used to solve many real-world problems. To achieve effective results in ANFIS training, there is a trend towards heuristic algorithms. Since the number of heuristic algorithms proposed in the literature is high, the field of study

Table 4 The terminology list of abbreviations used in this study

Abbreviation	Expansion
LSE	Least squares estimation
GD	Gradient descent
AI	Artificial intelligence
GA	Genetic algorithm
PSO	Particle swarm optimization
ABC	Artificial bee colony
ANN	Artificial neural network
BP	Backpropagation
HL	Hybrid learning
LM	Levenberg–Marquardt
KF	Kalman filter
EKF	Extended Kalman filter
SCKF	Square-root cubature Kalman filter
RLS	Recursive least squares
MLM	Modified Levenberg Marquardt
DE	Differential evolution
AIS	Artificial immune system
BA	Bee algorithm
CSO	Cat swarm optimization
CS	Cuckoo search
FA	Firefly algorithm
HS	Harmony search
MBA	Mine blast algorithm
SA	Simulated annealing
RCGA	Real-coded genetic algorithm
MOGA	Multi objective genetic algorithm
HTGA	Hybrid Taguchi-genetic algorithm
TBGA	Taguchi-based-genetic algorithm
GGA	Gradient genetic algorithm
IGA	Improved genetic algorithm
MGA	Modified genetic algorithm
SVD	Singular value decomposition
MPSO	Microhabitat particle swarm optimization
WT	Wavelet transform
EPSO	Evolutionary particle swarm optimization
FPSO	Fuzzy particle swarm optimization
HPSOWM	Hybrid particle swarm optimization with wavelet mutation
QPSO	Quantum-behaved particle swarm optimization algorithm
IQPSO	Improved quantum-behaved particle swarm optimization algorithm
IPSO	Improved particle swarm optimization

Table 4 continued

Abbreviation	Expansion
APSO	Accelerated particle swarm optimization
mPSO	Modified particle swarm optimization
aABC	Adaptive and hybrid artificial bee colony algorithm
AABC	Adaptive artificial bee colony algorithm
DEACS	Differential evolution with ant colony search
GBHS	Global-best harmony search

related to ANFIS training has shifted to this direction. Also, there is a risk of local minimum in derivative based methods. So, more effective performance is obtained by heuristics according to derivative based algorithms. Due to the reasons such as it, the use of heuristic algorithms has increased recently for training ANFIS. Standard versions and modifications of algorithms in heuristic based on ANFIS training have been used. As shown in Fig. 5; GA, PSO, ABC, DE, HS, CS, FA, SA, MBA and AIS are utilized in ANFIS training. Except HS, the standard versions of all other heuristics are used in ANFIS training. In addition, various modifications of heuristic algorithms such as GA, PSO, ABC, DE, HS and MBA are also performed. As described in Sect. 3.2, GA and PSO based methods have a significant place in ANFIS training. Although ABC algorithm is a quite new heuristic with respect to GA and PSO, it gains popularity in ANFIS training. It is very clear that studies are quite limited for other heuristic algorithms. From literature review, it can be seen that PSO, GA and ABC are quite successful in training ANFIS to solve real world problems. For that reason, these heuristic algorithms are in the top three ranks in ANFIS training.

In hybrid algorithms, while a heuristic algorithm is employed in the training of parameters related to membership functions, a derivative based method generally is utilized in training linear parameters. Hybrid algorithms achieve more successful results because they use the advantages of derivative and heuristic algorithms. In Fig. 6, hybrid algorithms are proposed. GA, PSO, DE, CS, BA and CSO algorithms are used in hybrid algorithm. It is seen that GA and PSO have important place especially. Against this, ABC, HS, FA, SA, MBA and AIS never have been used in hybrid algorithm.

When we look at the use of derivative based and heuristic algorithms in ANFIS training, it is highly possible that the methods to be proposed for this aim will be heuristic based due to its global search ability and flexibility. New studies are required to see the impact of other heuristics algorithms in ANFIS training. In addition to this, new hybrid training algorithms should be proposed. In the light of this information, it is anticipated that heuristic algorithm based training studies will continue and new effective methods will be presented in the following years. We hope that this survey will be very helpful for the researchers who want to study in ANFIS field.

References

- Abdullah AA, Xiong C, Zhang X, Kejia Z, Bachache NK (2014) Prediction and optimization approaches for modeling and selection of optimum machining parameters in CNC down milling operation. *Res J Appl Sci Eng Technol* 7:2908–2913
- Adnan MM, Sarkheyli A, Zain AM, Haron H (2015) Fuzzy logic for modeling machining process: a review. *Artif Intell Rev* 43:345–379

- Aguilar-Rivera R, Valenzuela-Rendón M, Rodríguez-Ortiz J (2015) Genetic algorithms and Darwinian approaches in financial applications: a survey. *Expert Syst Appl* 42:7684–7697
- Ahmadi MA, Ebadi M, Samadi A, Siuki MZ (2015) Phase equilibrium modeling of clathrate hydrates of carbon dioxide + 1,4-dioxine using intelligent approaches. *J Dispersion Sci Technol* 36:236–244. <https://doi.org/10.1080/01932691.2014.904792>
- Akachukwu CM, Aibinu AM, Nwohu MN, Salau HB (2014) A decade survey of engineering applications of genetic algorithm in power system optimization. In: 2014 5th international conference on intelligent systems, modelling and simulation (ISMS), IEEE, pp 38–42
- Akay B, Karaboga D (2015) A survey on the applications of artificial bee colony in signal, image, and video processing. *Signal Image Video Process* 9:967–990
- Akay B, Karaboga D (2011) Wavelet packets optimization using artificial bee colony algorithm. In: 2011 IEEE Congress on evolutionary computation (CEC), IEEE, pp 89–94
- Aliabadian Z, Sharifzadeh M, Sharafisafa M (2015) Optimizing the performance of ANFIS using the genetic algorithm to estimate the deformation modulus of rock mass. In: 49th US rock mechanics/geomechanics symposium 2015, pp 2233–2239
- Ali M, Ghatol A (2004) A neuro-fuzzy inference system for student modeling in web-based intelligent tutoring systems. In: Proceedings of international conference on cognitive systems, pp 14–19
- Allaoua B, Laoufi A, Gasbaoui B, Abderrahmani A (2009) Neuro-Fuzzy DC motor speed control using particle swarm optimization. *Leonardo Electron J Pract Technol* 8:1–18
- Almási A-D, Woźniak S, Cristea V, Leblebici Y, Engbersen T (2016) Review of advances in neural networks: neural design technology stack. *Neurocomputing* 174:31–41
- Aminifar S, Yosefi G (2007) Application of adaptive neuro fuzzy inference system (ANFIS) in implementing of new CMOS fuzzy logic controller (FLC) chip. In: AIP Conference Proceedings, pp 49–53. <https://doi.org/10.1063/1.2790190>
- Araghi S, Khosravi A, Creighton D (2014) ANFIS traffic signal controller for an isolated intersection. In: FCTA 2014 Proceedings of the international conference on fuzzy computation theory and applications, pp 175–180
- Araghi S, Khosravi A, Creighton D (2016) Design of an optimal ANFIS traffic signal controller by using Cuckoo search for an isolated intersection. In: Proceedings of 2015 IEEE international conference on systems, man, and cybernetics, SMC 2015, pp 2078–2083. <https://doi.org/10.1109/SMC.2015.363>
- Asadian A, Moshiri B, Sedigh AK, Lucas C (2005) Optimized data fusion in an intelligent integrated GPS/INS system using genetic algorithm. In: Proceedings WEC'05: 3rd World Enformatika conference, pp 221–224
- Ashuri B, Tavakolan M (2011) Fuzzy enabled hybrid genetic algorithm-particle swarm optimization approach to solve TCRO problems in construction project planning. *J Constr Eng Manag* 138:1065–1074
- Awadallah MA, Bayoumi EHE, Soliman HM (2009) Adaptive deadbeat controllers for brushless DC drives using PSO and ANFIS techniques. *J Electr Eng* 60:3–11
- Aydin M, Karakuzu C, Uçar M, Cengiz A, Çavuşlu MA (2013) Prediction of surface roughness and cutting zone temperature in dry turning processes of AISI304 stainless steel using ANFIS with PSO learning. *Int J Adv Manuf Technol* 67:957–967. <https://doi.org/10.1007/s00170-012-4540-2>
- Azadegan A, Porobic L, Ghazinoory S, Samouei P, Kheirkhah AS (2011) Fuzzy logic in manufacturing: a review of literature and a specialized application. *Int J Prod Econ* 132:258–270
- Azadeh A, Zarrin M (2016) An intelligent framework for productivity assessment and analysis of human resource from resilience engineering, motivational factors, HSE and ergonomics perspectives. *Saf Sci* 89:55–71. <https://doi.org/10.1016/j.ssci.2016.06.001>
- Babazadeh Khameneh N, Arabalibeik H, Salehian P, Setayeshi S (2012) Abnormal red blood cells detection using adaptive neuro-fuzzy system. In: Studies in health technology and informatics, pp 30–34. <https://doi.org/10.3233/978-1-61499-022-2-30>
- Baghban A, Bahadori M, Ahmad Z, Kashiwao T, Bahadori A (2016a) Modeling of true vapor pressure of petroleum products using ANFIS algorithm. *Pet Sci Technol* 34:933–939. <https://doi.org/10.1080/10916466.2016.1170843>
- Baghban A, Bahadori M, Rozyn J, Lee M, Abbas A, Bahadori A, Rahimali A (2016b) Estimation of air dew point temperature using computational intelligence schemes. *Appl Therm Eng* 93:1043–1052. <https://doi.org/10.1016/j.applthermaleng.2015.10.056>
- Bagheri A, Mohammadi Peyhani H, Akbari M (2014) Financial forecasting using ANFIS networks with quantum-behaved particle swarm optimization. *Expert Syst Appl* 41:6235–6250. <https://doi.org/10.1016/j.eswa.2014.04.003>
- Bagheri A, Nariman-Zadeh N, Jamali A, Dayjoori K (2009) Design of ANFIS networks using hybrid genetic and SVD method for the prediction of coastal wave impacts, vol 58

- Bagheri A, Zadeh NN, Haraj M, Moghaddam RY (2005) Identification of the dynamical parameters of a 2-R robot using ANFIS. In: IEEE international conference on mechatronics and automation, ICMA 2005, pp 505–509
- Bahadori A, Baghban A, Bahadori M, Lee M, Ahmad Z, Zare M, Abdollahi E (2016) Computational intelligent strategies to predict energy conservation benefits in excess air controlled gas-fired systems. *Appl Therm Eng* 102:432–446. <https://doi.org/10.1016/j.applthermaleng.2016.04.005>
- Bahamish HAA, Abdullah R (2010) Prediction of c-peptide structure using artificial bee colony algorithm. In: 2010 international symposium in information technology (ITSim), IEEE, pp 754–759
- Bakayani AE, Sahebi H, Ghiasi MM, Mirjordavi N, Esmailzadeh F, Lee M, Bahadori A (2016) Prediction of CO₂-oil molecular diffusion using adaptive neuro-fuzzy inference system and particle swarm optimization technique. *Fuel* 181:178–187. <https://doi.org/10.1016/j.fuel.2016.04.097>
- Barada S, Singh H (1998) Generating optimal adaptive fuzzy-neural models of dynamical systems with applications to control. *IEEE Trans Syst Man Cybern Part C Appl Rev* 28:371–391. <https://doi.org/10.1109/5326.704574>
- Barchi AC et al (2016) Artificial intelligence approach based on near-infrared spectral data for monitoring of solid-state fermentation. *Process Biochem* 51:1338–1347
- Basser H et al (2015) Hybrid ANFIS-PSO approach for predicting optimum parameters of a protective spur dike. *Appl Soft Comput J* 30:642–649. <https://doi.org/10.1016/j.asoc.2015.02.011>
- Begic Fazlic L, Avdagic A (2015) Prediction of lung nicotine concentration based on novel GA-ANFIS system approach. In: Proceedings of 2015 25th international conference on information, communication and automation technologies, ICAT 2015. <https://doi.org/10.1109/ICAT.2015.7340520>
- Begic Fazlic L, Avdagic A, Besic I (2015a) Prediction of heart attack risk using GA-ANFIS expert system prototype. In: PHealth 2015: proceedings of the 12th international conference on wearable micro and nano technologies for personalized health 2–4 June 2015 Västerås, Sweden, IOS Press, p 292
- Begic Fazlic L, Avdagic Z, Besic I (2015c) GA-ANFIS expert system prototype for detection of tar content in the manufacturing process. In: Proceedings of 2015 38th international convention on information and communication technology, electronics and microelectronics, MIPRO 2015, pp 1194–1199. <https://doi.org/10.1109/MIPRO.2015.7160457>
- Begic Fazlic L, Avdagic K, Omanovic S (2015b) GA-ANFIS expert system prototype for prediction of dermatological diseases. In: Studies in health technology and informatics, pp 622–626. <https://doi.org/10.3233/978-1-61499-512-8-622>
- Bhatt N, Chauhan NR (2015) Genetic algorithm applications on Job Shop Scheduling Problem: a review. In: 2015 international conference on soft computing techniques and implementations (ICSCTI), IEEE, pp 7–14
- Buyukyildiz M, Tezel G (2015) Utilization of PSO algorithm in estimation of water level change of Lake Beyşehir. *Theor Appl Climatol*. <https://doi.org/10.1007/s00704-015-1660-2>
- Cai CH, Du D, Liu ZY (2003) Battery state-of-charge (SOC) estimation using adaptive neuro-fuzzy inference system (ANFIS). In: IEEE International Conference on Fuzzy Systems, pp 1068–1073
- Cárdenas JJ, García A, Romeral JL, Kampouropoulos K (2011) Evolutive ANFIS training for energy load profile forecast for an IEMS in an automated factory. In: IEEE international conference on emerging technologies and factory automation, ETFA. <https://doi.org/10.1109/ETFA.2011.6059079>
- Carrano EG, Takahashi RHC, Caminhas WM, Neto OM (2008) A genetic algorithm for multiobjective training of ANFIS fuzzy networks. In: 2008 IEEE congress on evolutionary computation. CEC 2008, 3259–3265. <https://doi.org/10.1109/CEC.2008.4631239>
- Catalão JPS, Osório GJ, Pousinho HMI (2011a) Application of an intelligent system based on EPSO and ANFIS to price forecasting. In: 2011 16th international conference on intelligent system applications to power systems, ISAP 2011. <https://doi.org/10.1109/ISAP.2011.6082232>
- Catalão JPS, Osório GJ, Pousinho HMI (2011b) Short-term wind power forecasting using a hybrid evolutionary intelligent approach. In: 2011 16th international conference on intelligent system applications to power systems, ISAP 2011. <https://doi.org/10.1109/ISAP.2011.6082234>
- Catalão JPS, Pousinho HMI, Mendes VMF (2011c) Hybrid wavelet-PSO-ANFIS approach for short-term electricity prices forecasting. *IEEE Trans Power Syst* 26:137–144. <https://doi.org/10.1109/TPWRS.2010.2049385>
- Çavdar T (2016) PSO tuned ANFIS equalizer based on fuzzy C-means clustering algorithm. *AEU Int J Electron Commun* 70:799–807. <https://doi.org/10.1016/j.aeue.2016.03.006>
- Ch S, Mathur S (2010) Modeling uncertainty analysis in flow and solute transport model using adaptive neuro fuzzy inference system and particle swarm optimization. *KSCE J Civ Eng* 14:941–951. <https://doi.org/10.1007/s12205-010-0865-2>

- Chakrapani Y, Soundararajan K (2009) Adaptive neuro-fuzzy inference system based fractal image compression. Department of Electronic and Communications, JNTU College of Engineering, Hyderabad, p 2
- Chang BR, Tsai HF, Chen CM, Chang YS, Huang CF (2013a) Cloud-mobile computing based real-time VVoIP with PSO-ANFIS tuning. In: Proceedings of 2013 conference on technologies and applications of artificial intelligence, TAAI 2013, pp 115–121. <https://doi.org/10.1109/TAAI.2013.34>
- Chang BR, Tsai HF, Chen CM, Chang YS, Huang CF (2013b) PSO-ANFIS tuned real-time VVoIP in cloud with access control security. In: iFUZZY 2013—2013 international conference on fuzzy theory and its applications, pp 322–327. <https://doi.org/10.1109/iFuzzy.2013.6825458>
- Chang BR, Tsai HF, Wang YA, Chen CY (2016) Realization of secondary indexing to NoSQL database with intelligent adaptation. In: TAAI 2015—2015 conference on technologies and applications of artificial intelligence, pp 449–452. <https://doi.org/10.1109/TAAI.2015.7407078>
- Chang BR, Tsai HF, Wang YA, Kuo CF (2015) Intelligent adaptation to in-cloud NoSQL database remote backup between data centers. In: ACM international conference proceeding series. <https://doi.org/10.1145/2818869.2818892>
- Chang BR, Chen CM, Huang CF, Tsai HF (2014) Intelligent adaptation for UEC video/voice over IP with access control. *Int J Intell Inf Database Syst* 8:64–80. <https://doi.org/10.1504/IJIDS.2014.060463>
- Chao K-H (2014) An extension theory-based maximum power tracker using a particle swarm optimization algorithm. *Energy Convers Manag* 86:435–442
- Cheng CB, Lee ES (1999) Applying fuzzy adaptive network to fuzzy regression analysis. *Comput Math Appl* 38(2):123–140
- Chojaczyk A, Teixeira A, Neves L, Cardoso J, Soares CG (2015) Review and application of artificial neural networks models in reliability analysis of steel structures. *Struct Saf* 52:78–89
- Chy MI, Uddin MN (2009) Development and implementation of a new adaptive intelligent speed controller for IPMSM drive. *IEEE Trans Ind Appl* 45:1106–1115. <https://doi.org/10.1109/TIA.2009.2018918>
- Dai F, Kushida N, Shang L, Sugisaka M (2011) A survey of genetic algorithm-based face recognition. *Artif Life Robot* 16:271–274
- Dalkilic TE, Apaydin A (2014) Parameter estimation by ANFIS in cases where outputs are non-symmetric fuzzy numbers. *Int J Appl* 4(5)
- Dalkilic TE, Apaydin A (2009) A fuzzy adaptive network approach to parameter estimation in cases where independent variables come from an exponential distribution. *J Comput Appl Math* 233(1):36–45
- Dastranj MR, Ebrahimi E, Changizi N, Sameni E (2011) Control DC motorspeed with adaptive neuro-fuzzy control (ANFIS). *Aust J Basic Appl Sci* 5:1499–1504
- Dehghanian Serej A, Mojjallali H (2015) Speed control of elliptic motor using adaptive neural-fuzzy controller with on-line learning simulated under MATLAB/SIMULINK. In: *IntelliSys 2015—Proceedings of 2015 SAI intelligent systems conference*, pp 543–553. <https://doi.org/10.1109/IntelliSys.2015.7361193>
- Dewangan DN, Jha M, Qureshi MF, Banjare YP (2012) Real-time fault diagnostic and rectification system for bearing vibration of steam turbine by using adaptive neuro-fuzzy inference system and genetic algorithm—a novel approach. *Adv Model Anal B* 55:1–21
- Du Z, Li X, Mao Q (2015) A new online hybrid learning algorithm of adaptive neural fuzzy inference system for fault prediction. *Int J Model Identif Control* 23:68–76
- Entchev E, Yang L (2007) Application of adaptive neuro-fuzzy inference system techniques and artificial neural networks to predict solid oxide fuel cell performance in residential microgeneration installation. *J Power Sources* 170:122–129. <https://doi.org/10.1016/j.jpowsour.2007.04.015>
- Fang KL, Zhe W, Wei Z (2011) ANFIS-based fault diagnosis cloud model of oil parameter for automobile engine. In: *Proceedings 2011 international conference on mechatronic science, electric engineering and computer, MEC 2011*, pp 2458–2462. <https://doi.org/10.1109/MEC.2011.6025990>
- Fang H (2012) Adaptive neurofuzzy inference system in the application of the financial crisis forecast. *Int J Innov Manag Technol* 3:250
- Farid AM, Barakati SM (2014) DC Motor neuro-fuzzy controller using PSO identification. In: *22nd Iranian conference on electrical engineering, ICEE 2014*, pp 1162–1167. <https://doi.org/10.1109/IranianCEE.2014.6999711>
- Farid AM, Barakati SM, Seifipour N, Tayebi N (2013) Online ANFIS controller based on RBF identification and PSO. In: *2013 9th Asian control conference, ASCC 2013*. <https://doi.org/10.1109/ASCC.2013.6606232>
- Ganguly S, Sahoo N, Das D (2013) Multi-objective particle swarm optimization based on fuzzy-Pareto-dominance for possibilistic planning of electrical distribution systems incorporating distributed generation. *Fuzzy Sets Syst* 213:47–73

- Geetha G, Geethalakshmi SN (2012) Detecting epileptic seizures using electroencephalogram: a novel frequency domain feature extraction technique for seizure classification using fast ANFIS. In: ACM international conference proceeding series, pp 697–703. <https://doi.org/10.1145/2345396.2345510>
- Ghasemi E, Kalhori H, Bagherpour R (2016) A new hybrid ANFIS-PSO model for prediction of peak particle velocity due to bench blasting. *Eng Comput* 32:607–614. <https://doi.org/10.1007/s00366-016-0438-1>
- Göçken M, Boru A (2016) Integrating metaheuristics and ANFIS for daily mean temperature forecasting. *Int J Global Warm* 9:110–128. <https://doi.org/10.1504/IJGW.2016.074326>
- Gong Y, Qu Y (2012) Novel adaptive inverse control for permanent magnet synchronous motor servo system. *Przeglad Elektrotechniczny* 88:9–14
- Gong Y, Qu Y (2011) Adaptive inverse control based on MPSO–ANFIS for permanent magnet synchronous motor servo system. In: Proceedings—2011 3rd international conference on intelligent human-machine systems and cybernetics, IHMSC 2011, pp 173–176. <https://doi.org/10.1109/IHMSC.2011.48>
- Gunasekaran M, Ramaswami K (2011) A fusion model integrating anfis and artificial immune algorithm for forecasting indian stock market
- Guney K, Sarikaya N (2004a) Adaptive neuro-fuzzy inference system for the input resistance computation of rectangular microstrip antennas with thin and thick substrates. *J Electromagn Waves Appl* 18:23–39. <https://doi.org/10.1163/156939304322749599>
- Guney K, Sarikaya N (2004b) Computation of Resonant Frequency for Equilateral Triangular Microstrip Antennas Using the Adaptive Neuro-Fuzzy Inference System *Int J RF Microwave Compute-Aided Eng* 14:134–143. <https://doi.org/10.1002/mmce.10125>
- Guney K, Sarikaya N (2004c) Input resistance calculation for circular microstrip antennas using adaptive neuro-fuzzy inference system. *Int J Infrared Millimeter Waves* 25:703–716. <https://doi.org/10.1023/B:IJIM.0000020756.48454.31>
- Guney K, Sarikaya N (2007) Resonant frequency calculation for circular microstrip antennas with a dielectric cover using adaptive network-based fuzzy inference system optimized by various algorithms. *Prog Electromagn Res* 72:279–306. <https://doi.org/10.2528/PIER07031302>
- Guney K, Sarikaya N (2008a) Adaptive-network-based fuzzy inference system models for input resistance computation of circular microstrip antennas. *Microw Opt Technol Lett* 50:1253–1261. <https://doi.org/10.1002/mop.23354>
- Guney K, Sarikaya N (2008b) Adaptive-network-based fuzzy inference system models for narrow aperture dimension calculation of optimum gain pyramidal horns. *Neural Netw World* 18:341–363
- Guney K, Sarikaya N (2008c) Multiple adaptive-network-based fuzzy inference system for the synthesis of rectangular microstrip antennas with thin and thick substrates. *Int J RF Microw Comput Aided Eng* 18:359–375. <https://doi.org/10.1002/mmce.20294>
- Guney K, Sarikaya N (2009a) Comparison of adaptive-network-based fuzzy inference system models for resonant frequency computation of circular microstrip antennas. *J Commun Technol Electron* 54:369–380. <https://doi.org/10.1134/S1064226909040019>
- Guney K, Sarikaya N (2009b) Comparison of adaptive-network-based fuzzy inference systems for bandwidth calculation of rectangular microstrip antennas. *Expert Syst Appl* 36:3522–3535. <https://doi.org/10.1016/j.eswa.2008.02.008>
- Gupta NK, Tyagi N (2014) Neurofuzzy inference system for stage classified of diabetes. In: Souvenir of the 2014 IEEE international advance computing conference, IACC 2014, pp 1265–1269. <https://doi.org/10.1109/IAdCC.2014.6779509>
- Haghoosta T, Ismail WR (2015) Comparing typhoon intensity prediction with two different artificial intelligence models. *Evol Syst* 6:177–185. <https://doi.org/10.1007/s12530-014-9106-0>
- Hájek P, Olej V (2015) Intuitionistic fuzzy neural network: the case of credit scoring using text information, vol 517. https://doi.org/10.1007/978-3-319-23983-5_31
- Halim Z, Kalsoom R, Bashir S, Abbas G (2016) Artificial intelligence techniques for driving safety and vehicle crash prediction. *Artif Intell Rev* 46:351–387. <https://doi.org/10.1007/s10462-016-9467-9>
- Hasan AM, Samsudin K, Ramli AR (2011a) A novel intelligent predictor for low-rate global positioning system (GPS) system. *Sci Res Essays* 6:2348–2359
- Hasan AM, Samsudin K, Ramli AR (2011b) Optimizing of ANFIS for estimating INS error during GPS outages. *J Chin Inst Eng* 34:967–982. <https://doi.org/10.1080/02533839.2011.591970>
- Hassanein O, Anavatti SG, Ray T (2013) Black-box tool for nonlinear system identification based upon fuzzy system. *Int J Comput Intell Appl*. <https://doi.org/10.1142/S1469026813500090>
- Hassansin M, Taha MR, Noureldin A, El-Sheimy N (2004) Automization of an INS/GPS integrated system using genetic optimization. In: Automation Congress, 2004. Proceedings. World, IEEE, pp 347–352
- Ho WH, Chen JX, Lee IN, Su HC (2011) An ANFIS-based model for predicting adequacy of vancomycin regimen using improved genetic algorithm. *Expert Syst Appl* 38:13050–13056. <https://doi.org/10.1016/j.eswa.2011.04.109>

- Holland JH (1992) *Adaptation in natural and artificial systems: an introductory analysis with applications to biology, control, and artificial intelligence*. MIT press, Cambridge
- Hosseini SH, Etemadi AH (2008) Adaptive neuro-fuzzy inference system based automatic generation control. *Electr Power Syst Res* 78:1230–1239. <https://doi.org/10.1016/j.epsr.2007.10.007>
- Hussain K, Mohd Salleh MN, Leman AM (2016) Optimization of ANFIS using Mine Blast Algorithm for predicting strength of Malaysian small medium enterprises. In: 2015 12th international conference on fuzzy systems and knowledge discovery, FSKD 2015, pp 118–123. <https://doi.org/10.1109/FSKD.2015.7381926>
- Hussain K, Salleh MNM (2015) Optimization of fuzzy neural network using APSO for predicting strength of Malaysian SMEs. In: 2015 10th Asian control conference: emerging control techniques for a sustainable world. ASCC 2015, <https://doi.org/10.1109/ASCC.2015.7244638>
- Isazadeh G, Hooshmand RA, Khodabakhshian A (2012) Modeling and optimization of an adaptive dynamic load shedding using the ANFIS-PSO algorithm. *Simulation* 88:181–196. <https://doi.org/10.1177/0037549711400452>
- Isazadeh G, Khodabakhshian A, Gholipour E (2016) Optimal design of convertible static compensator supplementary damping controller to avoid wide area uncontrolled islanding. *IET Gener Transm Distrib* 10:2336–2350. <https://doi.org/10.1049/iet-gtd.2015.1025>
- Jamali A, Nariman-Zadeh N, Ashraf H, Jamali Z (2011) Robust Pareto design of ANFIS networks for non-linear systems with probabilistic uncertainties. In: INISTA 2011—2011 international symposium on innovations in intelligent systems and applications, pp 300–304. <https://doi.org/10.1109/INISTA.2011.5946080>
- Jang J-SR, Mizutani E (1996) Levenberg-Marquardt method for ANFIS learning. In: Biennial conference of the North American fuzzy information processing society—NAFIPS, pp 87–91
- Jang JSR (1993) ANFIS: adaptive-network-based fuzzy inference system. *IEEE Trans Syst Man Cybern* 23:665–685. <https://doi.org/10.1109/21.256541>
- Jannaty MH, Eghbalzadeh A, Hosseini SA (2015) Hybrid ANFIS model for predicting scour depth using particle swarm optimization. *Indian J Sci Technol*. <https://doi.org/10.17485/ijst/2015/v8i22/79321>
- Jarraya Y, Bouaziz S, Alimi AM, Abraham A (2014) Fuzzy modeling system based on hybrid evolutionary approach. In: 13th international conference on hybrid intelligent systems, HIS 2013, pp 72–77. <https://doi.org/10.1109/HIS.2013.6920457>
- Jhin C, Hwang KT (2015) Adaptive neuro-fuzzy inference system applied QSAR with quantum chemical descriptors for predicting radical scavenging activities of carotenoids. *PLoS ONE*. <https://doi.org/10.1371/journal.pone.0140154>
- Jiang HM, Kwong CK, Ip WH, Wong TC (2012) Modeling customer satisfaction for new product development using a PSO-based ANFIS approach. *Appl Soft Comput J* 12:726–734. <https://doi.org/10.1016/j.asoc.2011.10.020>
- Jiang H, Kwong CK, Law MC, Ip WH (2013) Development of customer satisfaction models for affective design using rough set and ANFIS approaches. *Proc Comput Sci*. <https://doi.org/10.1016/j.procs.2013.09.086>
- Jiang H, Kwong CK, Siu KWM, Liu Y (2015) Rough set and PSO-based ANFIS approaches to modeling customer satisfaction for affective product design. *Adv Eng Inform* 29:727–738. <https://doi.org/10.1016/j.aei.2015.07.005>
- Jurado F, Ortega M, Carpio J (2006) Power quality enhancement in fuel cells using genetic algorithms and ANFIS architecture. In: IEEE international symposium on industrial electronics, pp 757–762. <https://doi.org/10.1109/ISIE.2006.295729>
- Kabini K (2011) Review of ANFIS and its application in control of machining processes. *Sustain Res Innov Proc* 3
- Kar S, Das S, Ghosh PK (2014) Applications of neuro fuzzy systems: a brief review and future outline. *Appl Soft Comput* 15:243–259
- Karaboga D (2005) An idea based on honey bee swarm for numerical optimization. Technical report-tr06, Erciyes University, Engineering Faculty, Computer Engineering Department
- Karaboga N (2009) A new design method based on artificial bee colony algorithm for digital IIR filters. *J Franklin Inst* 346:328–348
- Karaboga D (2010) Artificial bee colony algorithm. *Scholarpedia* 5(3):6915
- Karaboga D, Basturk B (2007) A powerful and efficient algorithm for numerical function optimization: artificial bee colony (ABC) algorithm. *J Global Optim* 39:459–471
- Karaboga D, Kaya E (2016) An adaptive and hybrid artificial bee colony algorithm (aABC) for ANFIS training. *Appl Soft Comput J* 49:423–436. <https://doi.org/10.1016/j.asoc.2016.07.039>
- Karaboga D, Kaya E (2017) Training ANFIS by using the artificial bee colony algorithm. *Turk J Electr Eng Comput Sci* 25(3):1669–1679

- Karaboga D, Okdem S, Ozturk C (2012) Cluster based wireless sensor network routing using artificial bee colony algorithm. *Wireless Netw* 18:847–860
- Karaboga D, Gorkemli B, Ozturk C, Karaboga N (2014) A comprehensive survey: artificial bee colony (ABC) algorithm and applications. *Artif Intell Rev* 42:21–57
- Karaboga D, Akay B (2007) Artificial bee colony (ABC) algorithm on training artificial neural networks. In: *IEEE 15th signal processing and communications applications*, 2007. SIU 2007. IEEE, pp 1–4
- Karaboga D, Kaya E (2014) Training ANFIS using artificial bee colony algorithm for nonlinear dynamic systems identification. In: *Proceedings of 2014 22nd signal processing and communications applications conference*, SIU 2014, pp 493–496. <https://doi.org/10.1109/SIU.2014.6830273>
- Karaboga D, Training Kaya E (2013) ANFIS using artificial bee colony algorithm. In: *2013 IEEE international symposium on innovations in intelligent systems and applications*. IEEE INISTA 2013. <https://doi.org/10.1109/INISTA.2013.6577625>
- Karami A, Rezaei E, Rahimi M, Khani S (2013) Modeling of heat transfer in an air cooler equipped with classic twisted tape inserts using adaptive neuro-fuzzy inference system. *Chem Eng Commun* 200:532–542. <https://doi.org/10.1080/00986445.2012.715105>
- Karasulu B, Balli S (2010) Image segmentation using fuzzy logic, neural networks and genetic algorithms: survey and trends. *Mach Graph Vis Int J* 19:367–409
- Kasar MM, Bhattacharyya D, Kim T-h (2016) Face recognition using neural network: a review. *Int J Secur Appl* 10:81–100
- Kavitha S, Duraiswamy K (2011) Adaptive neuro-fuzzy inference system approach for the automatic screening of diabetic retinopathy in fundus images. *J Comput Sci* 7:1020–1026. <https://doi.org/10.3844/jcssp.2011.1020.1026>
- Kaya S, Guney K, Yildiz C, Turkmen M (2012) ANFIS models for synthesis of micro-coplanar stripline and asymmetric coplanar stripline with an infinitely wide strip. *Microw Opt Technol Lett* 54:460–467. <https://doi.org/10.1002/mop.26530>
- Kaya S, Guney K, Yildiz C, Turkmen M (2013) Anfis models for synthesis of open supported coplanar waveguides. *Neural Netw World* 23:553–569
- Kaya Y, Pehlivan H (2015) Feature selection using genetic algorithms for premature ventricular contraction classification. In: *2015 9th international conference on electrical and electronics engineering (ELECO)*, IEEE, pp 1229–1232
- Kennedy J, Eberhart R (1995) Particle swarm optimization. In: *IEEE international conference on neural networks—Conference Proceedings*, pp 1942–1948
- Khalid HM, Rizvi SZ, Cheded L, Doraiswami R, Khoukhi A (2010) A PSO-trained adaptive neuro-fuzzy inference system for fault classification. In: *Proceedings of the international conference on fuzzy computation and international conference on neural computation*, ICFC and ICNC 2010, pp 399–405
- Khatibinia M, Salajegheh J, Fadaee MJ, Salajegheh E (2012) Prediction of failure probability for soilstructure interaction system using modified ANFIS by hybrid of FCM-FPSO. *Asian J Civ Eng* 13:1–27
- Khim Chong C, Saberi Mohamad M, Deris S, Shahir Shamsir M, Wen Choon Y, En Chai L (2014) A review on modelling methods, pathway simulation software and recent development on differential evolution algorithms for metabolic pathways in systems biology. *Curr Bioinf* 9:509–521
- Khoshbin F, Bonakdari H, Ashraf Taleh SH, Ebtehaj I, Zaji AH, Azimi H (2016) Adaptive neuro-fuzzy inference system multi-objective optimization using the genetic algorithm/singular value decomposition method for modelling the discharge coefficient in rectangular sharp-crested side weirs. *Eng Optim* 48:933–948. <https://doi.org/10.1080/0305215X.2015.1071807>
- Khosravi A, Nahavandi S, Creighton D (2011) Prediction interval construction and optimization for adaptive neurofuzzy inference systems. *IEEE Trans Fuzzy Syst* 19:983–988. <https://doi.org/10.1109/TFUZZ.2011.2130529>
- Kockanat S, Karaboga N (2015a) The design approaches of two-dimensional digital filters based on meta-heuristic optimization algorithms: a review of the literature. *Artif Intell Rev* 44:265–287
- Kockanat S, Karaboga N (2015b) A novel 2D-ABC adaptive filter algorithm: a comparative study. *Digit Signal Proc* 40:140–153
- Koukol M, Zajíčková L, Marek L, Tuček P (2015) Fuzzy logic in traffic engineering: a review on signal control. *Math Probl Eng* 2015
- Koza T, Karaboga N (2017) Quadrature mirror filter bank design for mitral valve doppler signal using artificial bee colony algorithm. *Elektron Elektrotech* 23(1):57–62
- Liao Z, Zhang F (2013) Identification of nonlinear system based on ANFIS with Hybrid fuzzy clustering. *Inf Technol J* 12:8349–8353. <https://doi.org/10.3923/itj.2013.8349.8353>
- Lin X, Sun J, Palade V, Fang W, Wu X, Xu W (2012) Training ANFIS parameters with a quantum-behaved particle swarm optimization algorithm, vol 7331 LNCS. https://doi.org/10.1007/978-3-642-30976-2_18

- Liu H, Huang M, Yoo CK (2013a) A fuzzy neural network-based soft sensor for modeling nutrient removal mechanism in a full-scale wastewater treatment system. *Desalin Water Treat* 51:6184–6193. <https://doi.org/10.1080/19443994.2013.780757>
- Liu P, Leng W, Fang W (2013) Training ANFIS model with an improved quantum-behaved particle swarm optimization algorithm. *Math Probl Eng*. <https://doi.org/10.1155/2013/595639>
- Liu Z, Wang X (2012) A PSO-based algorithm for load balancing in virtual machines of cloud computing environment. *Adv Swarm Intell* 142–147
- Lochan K, Roy B (2015) Control of two-link 2-DOF robot manipulator using fuzzy logic techniques: a review. In: *Proceedings of fourth international conference on soft computing for problem solving*. Springer, pp 499–511
- Loukas YL (2001) Adaptive neuro-fuzzy inference system: An instant and architecture-free predictor for improved QSAR studies. *J Med Chem* 44:2772–2783. <https://doi.org/10.1021/jm000226c>
- Lu JT, Chang YC, Ho CY (2015) The optimization of chiller loading by adaptive neuro-fuzzy inference system and genetic algorithms. *Math Probl Eng*. <https://doi.org/10.1155/2015/306401>
- Lutfy OF, Mohd Noor SB, Marhaban MH (2011a) A genetically trained simplified ANFIS controller to control nonlinear MIMO systems. In: *International conference on electrical, control and computer engineering, InECCE 2011*, pp 349–354. <https://doi.org/10.1109/INECCE.2011.5953905>
- Lutfy OF, Mohd Noor SB, Marhaban MH, Abbas KA (2009) A genetically trained adaptive neuro-fuzzy inference system network utilized as a proportional–integral–derivative-like feedback controller for nonlinear systems. *Proc Inst Mech Eng Part I J Syst Control Eng* 223:309–321. <https://doi.org/10.1243/09596518JSCE683>
- Lutfy OF, Mohd Noor SB, Marhaban MH, Abbas KA (2010a) A Simplified PID-like ANFIS controller trained by genetic algorithm to control nonlinear systems. *Aust J Basic Appl Sci* 4:6331–6345
- Lutfy OF, Mohd Noor SB, Marhaban MH, Abbas KA (2010b) Utilizing global-best harmony search to train a PID-like ANFIS controller. *Aust J Basic Appl Sci* 4:6319–6330
- Lutfy OF, Mohd Noor SB, Marhaban MH (2011b) A simplified adaptive neuro-fuzzy inference system (ANFIS) controller trained by genetic algorithm to control nonlinear multi-input multi-output systems. *Sci Res Essays* 6:6475–6486
- Lutfy OF, Mohd Noor SB, Marhaban MH, Abbas KA (2011c) Non-linear modelling and control of a conveyor-belt grain dryer utilizing neuro-fuzzy systems. *Proc Inst Mech Eng Part I J Syst Control Eng* 225:611–622. <https://doi.org/10.1177/2041304110394559>
- Mahapatra S, Daniel R, Dey DN, Nayak SK (2015) Induction motor control using PSO-ANFIS. *Proc Comput Sci*. <https://doi.org/10.1016/j.procs.2015.04.212>
- Maji K, Pratihari DK, Nath AK (2013) Analysis and synthesis of laser forming process using neural networks and neuro-fuzzy inference system. *Soft Comput* 17:849–865. <https://doi.org/10.1007/s00500-012-0949-7>
- Malarvizhi K, Kiruba R, Kumar M, Brindha M (2014) Estimation of ANFIS parameters for a non linear system using extended kalman filter and particle swarm optimization. *Int J Appl Eng Res* 9:10363–10374
- Mansour N, Kanj F, Khachfe H (2012) Particle swarm optimization approach for protein structure prediction in the 3D HP model Interdisciplinary sciences. *Comput Life Sci* 4:190
- Marzbanrad J, Jamali A (2009) Design of ANFIS networks using hybrid genetic and SVD methods for modeling and prediction of rubber engine mount stiffness. *Int J Autom Technol* 10:167–174. <https://doi.org/10.1007/s12239-009-0020-6>
- Marzi H, Haj Darwish A, Helfawi H (2016) Training ANFIS using the enhanced bees algorithm and least squares estimation. *Intell Autom Soft Comput*. <https://doi.org/10.1080/10798587.2016.1196880>
- Mehrdad A, Nariman-Zadeh N, Jamali A, Teymoorzadeh A (2005) ANFIS networks design using hybrid genetic and SVD methods for modelling of the level variations of the Caspian Sea. *WSEAS Trans Inf Sci Appl* 2:121–126
- Ming L, Hai H, Aimin Z, Yingde S, Zhao L, Xingguo Z (2012) Modeling of mechanical properties of as-cast Mg–Li–Al alloys based on PSO-BP algorithm. *Res Dev* 9(2)
- Mohamed S, Hameed I (2015) A GA-based adaptive neuro-fuzzy controller for greenhouse climate control system. *Alexandria Eng J* <https://doi.org/10.1016/j.aej.2014.04.009>
- Mohanty PK, Parhi DR (2015) A new hybrid optimization algorithm for multiple mobile robots navigation based on the CS-ANFIS approach. *Memet Comput* 7:255–273. <https://doi.org/10.1007/s12293-015-0160-3>
- Mohd Salleh MN, Hussain K (2016) Accelerated mine blast algorithm for ANFIS training for solving classification problems. *Int J Softw Eng Appl* 10:161–168. <https://doi.org/10.14257/ijseia.2016.10.6.13>
- Mollaiy-Berneti S (2016) Determination of minimum miscibility pressure in CO₂-IOR projects with the aid of hybrid neuro-fuzzy system. *Fuel* 180:59–70. <https://doi.org/10.1016/j.fuel.2016.04.015>

- Monirvaghefi H, Rafiee Sandgani M, Aliyari Shoorehdeli M (2013) Interval type-2 adaptive network-based fuzzy inference system (ANFIS) with Type-2 non-singleton fuzzification. In: 13th Iranian conference on fuzzy systems, IFSC 2013. <https://doi.org/10.1109/IFSC.2013.6675612>
- Mustafa MW, Mustapha M, Khalid SN, Abubakar I (2016) Wavelet-based short-term load forecasting using optimized anfis. *ARPN J Eng Appl Sci* 11:6920–6927
- Nariman-Zadeh N, Darvizeh A, Dadfarmai MH (2003) Adaptive neurofuzzy inference systems networks design using hybrid genetic and singular value decomposition methods for modeling and prediction of the explosive cutting process. *Artif Intell Eng Des Anal Manuf AIEDAM* 17:313–324
- Nariman-Zadeh N, Darvizeh A, Dadfarmai MH (2004) Design of ANFIS networks using hybrid genetic and SVD methods for the modelling of explosive cutting process. *J Mater Process Technol* 155–156:1415–1421. <https://doi.org/10.1016/j.jmatprotec.2004.04.228>
- Nasiri M, Faez K (2012) Extracting fetal electrocardiogram signal using ANFIS trained by genetic algorithm. In: 2012 international conference on biomedical engineering, ICoBE 2012, pp 197–202. <https://doi.org/10.1109/ICoBE.2012.6179004>
- Nasiri M, Faez K, Nasrabadi AM (2011) A new method for extraction of fetal electrocardiogram signal based on adaptive nero-fuzzy inference system. In: 2011 IEEE international conference on signal and image processing applications, ICSIPA 2011, pp 456–461. <https://doi.org/10.1109/ICSIPA.2011.6144151>
- Nhu HN, Nitsuwat S, Sodanil M (2013) Prediction of stock price using an adaptive Neuro-Fuzzy Inference System trained by Firefly Algorithm. In: 2013 International Computer Science and Engineering Conference, ICSEC 2013, 302–307. <https://doi.org/10.1109/ICSEC.2013.6694798>
- Oh SK, Park KJ, Pedrycz W (2006) Consecutive identification of ANFIS-based fuzzy systems with the aid of genetic data granulation, vol 3972 LNCS. https://doi.org/10.1007/11760023_121
- Okdem S, Karaboga D, Ozturk C (2011) An application of wireless sensor network routing based on artificial bee colony algorithm. In: 2011 IEEE Congress on evolutionary computation (CEC). IEEE, pp 326–330
- Oliveira MV, Schirru R (2009) Applying particle swarm optimization algorithm for tuning a neuro-fuzzy inference system for sensor monitoring. *Prog Nucl Energy* 51:177–183. <https://doi.org/10.1016/j.pnucene.2008.03.007>
- Orouskhani M, Mansouri M, Orouskhani Y, Teshnehlab M (2013) A hybrid method of modified cat swarm optimization and gradient descent algorithm for training ANFIS. *Int J Comput Intell Appl*. <https://doi.org/10.1142/S1469026813500077>
- Osório GJ, Matias JCO, Pousinho HMI, Catalão JPS (2012a) Hybrid neuro-fuzzy evolutionary approach for short-term wind power forecasting. In: Proceedings of the Mediterranean electrotechnical conference—MELECON, pp 257–260. <https://doi.org/10.1109/MELCON.2012.6196427>
- Osório GJ, Pousinho HMI, Matias JCO, Catalão JPS (2012b) Intelligent approach for forecasting in power engineering systems. In: Proceedings of IEEE 16th international conference on intelligent engineering systems, INES 2012, pp 297–302. <https://doi.org/10.1109/INES.2012.6249848>
- Ozdemir G, Karaboga N (2017) A review on the cosine modulated filter bank studies using meta-heuristic optimization algorithms. *Artif Intell Rev* 1–25
- Pamućar D, Ljubojević S, Kostadinović D, Đorović B (2016) Cost and risk aggregation in multi-objective route planning for hazardous materials transportation—a neuro-fuzzy and artificial bee colony approach. *Expert Syst Appl* 65:1–15. <https://doi.org/10.1016/j.eswa.2016.08.024>
- Pan WT (2009) Forecasting classification of operating performance of enterprises by zscore combining ANFIS and genetic algorithm. *Neural Comput Appl* 18:1005–1011. <https://doi.org/10.1007/s00521-009-0243-5>
- Pandey SK, Mohanty SR, Kishor N, Catalão JP (2014) Frequency regulation in hybrid power systems using particle swarm optimization and linear matrix inequalities based robust controller design. *Int J Electr Power Energy Syst* 63:887–900
- Patil S, Mandal S, Hegde A, Alavandar S (2011) Neuro-fuzzy based approach for wave transmission prediction of horizontally interlaced multilayer moored floating pipe breakwater. *Ocean Eng* 38:186–196
- Plagianakos V, Tasoulis D, Vrahatis M (2008) A review of major application areas of differential evolution. In: *Advances in differential evolution*. Springer, pp 197–238
- Pousinho HMI, Catalão JPS, Mendes VMF (2010) Wind power short-term prediction by a hybrid PSO-ANFIS approach. In: Proceedings of the Mediterranean electrotechnical conference—MELECON, pp 955–960. <https://doi.org/10.1109/MELCON.2010.5475923>
- Pousinho HMI, Mendes VMF, Catalão JPDS (2011) A hybrid PSO-ANFIS approach for short-term wind power prediction in Portugal. *Energy Convers Manag* 52:397–402
- Pousinho HMI, Mendes VMF, Catalão JPS (2012) Short-term electricity prices forecasting in a competitive market by a hybrid PSO-ANFIS approach. *Int J Electr Power Energy Syst* 39:29–35. <https://doi.org/10.1016/j.jepes.2012.01.001>

- Premkumar K, Manikandan BV (2015) Fuzzy PID supervised online ANFIS based speed controller for brushless DC motor. *Neurocomputing* 157:76–90. <https://doi.org/10.1016/j.neucom.2015.01.032>
- Priyadharsini SS, Rajan SE (2014a) An Efficient method for the removal of ECG artifact from measured EEG Signal using PSO algorithm. *Int J Advance Soft Comput Appl* 6
- Priyadharsini SS, Rajan SE (2014b) Evolutionary computing based approach for the removal of ECG artifact from the corrupted EEG signal. *Technol Health Care* 22:835–846. <https://doi.org/10.3233/THC-140860>
- Qi Z, Zhu X, Cao G (2006) Temperature modeling and control of direct methanol fuel cell based on adaptive neural fuzzy technology. *High Technol Lett* 12:421–426
- Rai AA, Pai PS, Rao BRS (2015) Prediction models for performance and emissions of a dual fuel CI engine using ANFIS. *Sadhana Acad Proc Eng Sci* 40:515–535. <https://doi.org/10.1007/s12046-014-0320-z>
- Rajasekaran MP, Meena RS (2012) Application of adaptive neuro-fuzzy inference systems for MR image classification and tumour detection. *Int J Biomed Eng Technol* 9:133–146. <https://doi.org/10.1504/IJBET.2012.047746>
- Rastegar F, Araabi BN, Lucas C (2005) An evolutionary fuzzy modeling approach for ANFIS architecture. In: 2005 IEEE Congress on evolutionary computation, IEEE CEC 2005. Proceedings, pp 2182–2189
- Rezazadeh S, Ardestani MA, Sadeghi PS (2014) Optimal attitude control of a quadrotor UAV using adaptive neuro-fuzzy inference system (ANFIS). In: 3rd international conference on control, instrumentation, and automation, ICCIA 2013, pp 219–223. <https://doi.org/10.1109/ICCIAutom.2013.6912838>
- Rezvani A, Izadbakhsh M, Gandomkar M, Vafaei S (2015) Implementing GA-ANFIS for maximum power point tracking in PV system. *Indian J Sci Technol* 8:982–991. <https://doi.org/10.17485/ijst/2015/v8i10/51832>
- Rini DP, Shamsuddin SM, Yuhani SS (2013) Balanced the trade-offs problem of ANFIS using particle swarm optimization. *Telkomnika* 11:611–616. <https://doi.org/10.12928/TELKOMNIKA.v11i3.1271>
- Rini DP, Shamsuddin SM, Yuhani SS (2016) Particle swarm optimization for ANFIS interpretability and accuracy. *Soft Comput* 20:251–262. <https://doi.org/10.1007/s00500-014-1498-z>
- Salleh MNM, Hussain K (2016) A review of training methods of ANFIS for applications in business and economics. *Int J u-e-Serv Sci Technol* 9:165–172
- Salmalian K, Soleimani M (2011) Modelling of energy absorption in square cross-section aluminum energy absorbers by hybrid ANFIS networks. *Int J Math Models Methods Appl Sci* 5:1154–1161
- Samanta B (2005) Machine fault detection using neuro-fuzzy inference system and genetic algorithms. In: ASME 2005 international design engineering technical conferences and computers and information in engineering conference. American Society of Mechanical Engineers, pp 1031–1038
- San PP, Ling SH, Nguyen HT (2012) Intelligent detection of hypoglycemic episodes in children with type 1 diabetes using adaptive neural-fuzzy inference system. In: Proceedings of the annual international conference of the IEEE engineering in medicine and biology society, EMBS, pp 6325–6328. <https://doi.org/10.1109/EMBC.2012.6347440>
- Saradhadevi V, Sundaram V (2012) An enhanced two-stage impulse noise removal technique for sar images based on fast ANFIS and fuzzy decision. *Eur J Sci Res* 68:506–522
- Sargolzaei A, Faez K, Sargolzaei S (2011) A new method for Foetal Electrocardiogram extraction using adaptive neuro-fuzzy inference system trained with PSO algorithm. In: IEEE international conference on electro information technology. <https://doi.org/10.1109/EIT.2011.5978624>
- Sarıca B, Eğrioğlu E, Aşıkil B (2016) A new hybrid method for time series forecasting: AR-ANFIS. *Neural Comput Appl*. <https://doi.org/10.1007/s00521-016-2475-5>
- Sarkheyli A, Zain AM, Sharif S (2015a) A multi-performance prediction model based on ANFIS and new modified-GA for machining processes. *J Intell Manuf* 26:703–716. <https://doi.org/10.1007/s10845-013-0828-9>
- Sarkheyli A, Zain AM, Sharif S (2015b) Robust optimization of ANFIS based on a new modified GA. *Neurocomputing* 166:357–366. <https://doi.org/10.1016/j.neucom.2015.03.060>
- Seydi Ghomsheh V, Aliyari Shoorehdeli M, Teshnehlab M (2007) Training ANFIS structure with modified PSO algorithm. In: 2007 Mediterranean conference on control and automation, MED, <https://doi.org/10.1109/MED.2007.4433927>
- Shamsipour M, Pahlevani Z, Shabani MO, Mazahery A (2016) Optimization of the EMS process parameters in compocasting of high-wear-resistant Al-nano-TiC composites. *Appl Phys A Mater Sci Process*. <https://doi.org/10.1007/s00339-016-9840-1>
- Sharma S, Kalra U, Srivathsan S, Rana KPS, Kumar V (2015) Efficient air pollutants prediction using ANFIS trained by modified PSO algorithm. In: 2015 4th international conference on reliability, infocom technologies and optimization: trends and future directions, ICRITO 2015. <https://doi.org/10.1109/ICRITO.2015.7359316>
- Sheeja Agustin A, Suresh Babu S (2014) An improved thyroid tumor segmentation and classification approach using ANFIS-AABC. *Int J Appl Eng Res* 9:13387–13408

- Sheeja Agustin A, Suresh Babu S (2015) Tissue classification and boundary based segmentation in thyroid ultrasound images. *Int J Appl Eng Res* 10:21565–21581
- Sheniha SF, Priyadharsini SS, Rajan SE (2013) Removal of artifact from EEG signal using differential evolution algorithm. In: International conference on communication and signal processing, ICCSP 2013. Proceedings, pp 134–138. <https://doi.org/10.1109/iccsp.2013.6577031>
- Shoorehdeli MA, Teshnehlal M, Sedigh AK (2006) A novel training algorithm in ANFIS structure. In: Proceedings of the American control conference, pp 5059–5064
- Shoorehdeli MA, Teshnehlal M, Sedigh AK (2007) Novel hybrid learning algorithms for tuning ANFIS parameters using adaptive weighted PSO. In: IEEE international conference on fuzzy systems. <https://doi.org/10.1109/FUZZY.2007.4295571>
- Shoorehdeli M, Teshnehlal M, Sedigh AK (2009a) Identification using ANFIS with intelligent hybrid stable learning algorithm approaches. *Neural Comput Appl* 18:157–174. <https://doi.org/10.1007/s00521-007-0168-9>
- Shoorehdeli MA, Teshnehlal M, Sedigh AK (2009b) Training ANFIS as an identifier with intelligent hybrid stable learning algorithm based on particle swarm optimization and extended Kalman filter. *Fuzzy Sets Syst* 160:922–948. <https://doi.org/10.1016/j.fss.2008.09.011>
- Shoorehdeli MA, Teshnehlal M, Sedigh AK, Khanesar MA (2009c) Identification using ANFIS with intelligent hybrid stable learning algorithm approaches and stability analysis of training methods. *Appl Soft Comput J* 9:833–850. <https://doi.org/10.1016/j.asoc.2008.11.001>
- Sindhiya S, Gunasundari S (2014) A survey on genetic algorithm based feature selection for disease diagnosis system. In: 2014 international conference on computer communication and systems, IEEE, pp 164–169
- Soto J, Melin P (2014) Genetic optimization of type-2 fuzzy integrators in ensembles of ANFIS models for time series prediction, vol 547. https://doi.org/10.1007/978-3-319-05170-3_6
- Soto J, Melin P (2015) Optimization of the interval type-2 fuzzy integrators in ensembles of ANFIS models for time series prediction: case of the Mexican stock exchange, vol 601. https://doi.org/10.1007/978-3-319-17747-2_3
- Soto J, Melin P, Castillo O (2013) Optimization of interval type-2 and type-1 fuzzy integrators in ensembles of ANFIS models with genetic algorithms. In: 2013 World congress on nature and biologically inspired computing. NaBIC 2013, pp 41–46. <https://doi.org/10.1109/NaBIC.2013.6617876>
- Soto J, Melin P, Castillo O (2015a) Optimization of the type-1 and interval type-2 fuzzy integrators in Ensembles of ANFIS models for prediction of the Dow Jones time series. In: IEEE SSCI 2014 - 2014 IEEE symposium series on computational intelligence - CIDM 2014: 2014 IEEE symposium on computational intelligence and data mining, proceedings, pp 186–193. <https://doi.org/10.1109/CIDM.2014.7008666>
- Soto J, Melin P, Castillo O (2015b) Time series prediction using ensembles of ANFIS models with particle swarm optimization of the fuzzy integrators, vol 9413. https://doi.org/10.1007/978-3-319-27060-9_39
- Storn R, Price K (1997) Differential evolution—a simple and efficient heuristic for global optimization over continuous spaces. *J Global Optim* 11:341–359
- Suganthi L, Iniyan S, Samuel AA (2015) Applications of fuzzy logic in renewable energy systems—a review. *Renew Sustain Energy Rev* 48:585–607. <https://doi.org/10.1016/j.rser.2015.04.037>
- Suja Priyadharsini S, Edward Rajan S, Femilin Sheniha S (2016) A novel approach for the elimination of artefacts from EEG signals employing an improved artificial immune system algorithm. *J Exp Theor Artif Intell* 28:239–259. <https://doi.org/10.1080/0952813X.2015.1020571>
- Suja KR, Raglend IJ (2012) Genetic algorithm-neuro-fuzzy controller (GANFC) based UPQC controller for compensating PQ problem. *Eur J Sci Res* 78:184–197
- Suman S, Khan S, Das S, Chand S (2016) Slope stability analysis using artificial intelligence techniques. *Nat Hazards* 84:727–748
- Tabesh M, Dini M (2009) Fuzzy and neuro-fuzzy models for short-term water demand forecasting in Tehran. *Iran J Sci Technol, Transaction B: Eng* 33:61–77
- Taghavifar H, Mardani A (2014) On the modeling of energy efficiency indices of agricultural tractor driving wheels applying adaptive neuro-fuzzy inference system *J Terramech* 56:37–47. <https://doi.org/10.1016/j.jterra.2014.08.002>
- Tavoosi J, Suratgar AA, Menhaj MB (2016) Stable ANFIS2 for nonlinear system identification. *Neurocomputing* 182:235–246. <https://doi.org/10.1016/j.neucom.2015.12.030>
- Teshnehlal M, Shoorehdeli MA, Sedigh AK (2008) Novel hybrid learning algorithms for tuning ANFIS parameters as an identifier using fuzzy PSO. In: Proceedings of 2008 IEEE international conference on networking, sensing and control, ICNSC, pp 111–116. <https://doi.org/10.1109/ICNSC.2008.4525193>
- Thakral P, Arora V, Kukreti S, Bakhshi A (2013) In-silico engineering of intrinsically conducting copolymers using particle swarm optimization algorithm. *Indian J Chem* 52(A): 317–326

- Toha SF, Tokhi MO (2010) ANFIS modelling of a twin rotor system using particle swarm optimisation and RLS. In: 2010 IEEE 9th international conference on cybernetic intelligent systems, CIS 2010. <https://doi.org/10.1109/UKRICIS.2010.5898130>
- Topalov AV, Kayacan E, Oniz Y, Kaynak O (2009) Adaptive neuro-fuzzy control with sliding mode learning algorithm: application to antilock braking system. In: Asian Control Conference, 2009. ASCC 2009. 7th, IEEE, pp 784–789
- Torres-Salomao LA, Anzures-Marin J, Orozco-Sixtos JM, Ramírez-Zavala S (2015) ANFIS data driven modeling and real-time fuzzy Logic Controller test for a two tanks hydraulic system. In: 2015 IEEE international conference on evolving and adaptive intelligent systems, EAIS 2015. <https://doi.org/10.1109/EAIS.2015.7368778>
- Tsai JT, Chiu KY, Chou JH (2015a) Optimal design of SAW gas sensing device by using improved adaptive neuro-fuzzy inference system. IEEE Access 3:420–429. <https://doi.org/10.1109/ACCESS.2015.2427291>
- Tsai JT, Chou JH, Lin CF (2015b) Designing micro-structure parameters for backlight modules by using improved adaptive neuro-fuzzy inference system. IEEE Access 3:2626–2636. <https://doi.org/10.1109/ACCESS.2015.2508144>
- Turki M, Bouzaida S, Sakly A, M'Sahli F (2012) Adaptive control of nonlinear system using neuro-fuzzy learning by PSO algorithm. In: Proceedings of the Mediterranean electrotechnical conference—MELECON, pp 519–523. <https://doi.org/10.1109/MELCON.2012.6196486>
- Turkmen M, Yildiz C, Guney K, Kaya S (2009) Comparison of adaptive-network-based fuzzy inference system models for analysis of conductor-backed asymmetric coplanar waveguides. Prog Electromagn Res M 8:1–13. <https://doi.org/10.2528/PIERM09050803>
- Turkmen M, Yildiz C, Guney K, Kaya S (2010a) Adaptive-network-based fuzzy inference system models for computing the characteristic impedances of air-suspended trapezoidal and rectangular-shaped microshield lines. Microw Opt Technol Lett 52:20–24. <https://doi.org/10.1002/mop.24829>
- Turkmen M, Yildiz C, Guney K, Kaya S (2010b) Anfis models for the quasistatic analysis of coplanar strip line structures. Microw Opt Technol Lett 52:1990–1996. <https://doi.org/10.1002/mop.25372>
- Vadood M (2014) Predicting the color index of acrylic fiber using fuzzy-genetic approach. J Text Inst 105:779–788. <https://doi.org/10.1080/00405000.2013.849844>
- Vasilakos AV, Tang Y, Yao Y (2016) Neural networks for computer-aided diagnosis in medicine: a review. Neurocomputing 216:700–708
- Vieira J, Dias FM, Mota A (2004) Neuro-fuzzy systems: a survey. In: 5th WSEAS NNA international conference
- Vijayalakshmi S, Girish G (2015) Artificial neural networks for spot electricity price forecasting: a review. Int J Energy Econ Policy 5:1092–1097
- Walia N, Kumar S, Singh H (2015a) A survey on applications of adaptive neuro fuzzy inference system. Int J Hybrid Inf Technol 8:343–350
- Walia N, Singh H, Sharma A (2015b) ANFIS: adaptive neuro-fuzzy inference system—a survey. Int J Comput Appl 123:32–38
- Wang JS (2007) Parameters optimization of ANFIS based on particle swarm optimization (Shiyou Huangong Gaodeng Xuexiao Xuebao). J Pet Univ 20:41–44
- Wang JS, Ning CX (2015) ANFIS based time series prediction method of bank cash flow optimized by adaptive population activity PSO algorithm. Information 6:300–313. <https://doi.org/10.3390/info6030300> (Switzerland)
- Wang JN, Shen QT, Chen XZ (2006) Evolutionary design of adaptive neuro-fuzzy inference system based on hybrid cooperative particle swarm optimization (Xitong Gongcheng Lilun Shijian). Syst Eng Theory Pract 26:48–54
- Wang R, Zhang J, Zhang Y, Wang X (2012b) Assessment of human operator functional state using a novel differential evolution optimization based adaptive fuzzy model. Biomed Signal Process Control 7:490–498. <https://doi.org/10.1016/j.bspc.2011.09.004>
- Wang X, Hu H, Zhang A (2014) Concentration measurement of three-phase flow based on multi-sensor data fusion using adaptive fuzzy inference system. Flow Meas Instrum 39:1–8. <https://doi.org/10.1016/j.flowmeasinst.2014.04.003>
- Wang J, Gao XZ, Tanskanen JMA, Guo P (2012a) Epileptic EEG signal classification with ANFIS based on harmony search method. In: Proceedings of the 2012 8th international conference on computational intelligence and security, CIS 2012, pp 690–694. <https://doi.org/10.1109/CIS.2012.159>
- Wanglei, Shao P (2016) Intelligent control in smart home based on adaptive neuro fuzzy inference system. In: Proceedings of 2015 Chinese automation congress, CAC 2015, pp 1154–1158. <https://doi.org/10.1109/CAC.2015.7382672>

- Wang J, Ma L, Xu Y, Li L (2011) ANFIS indoor positioning system based on improved-GA in WLAN environment. In: Proceedings of 2011 3rd international conference on intelligent human-machine systems and cybernetics, IHMSC 2011, pp 147–151. <https://doi.org/10.1109/IHMSC.2011.106>
- Wei FM, Zhang JP, Li B, Yang J (2014) A survey of quantum genetic algorithm for combinatorial optimization problems. In: Applied mechanics and materials, Trans Tech Publ, pp 822–826
- Wu Q, Cai W, Wang X, Shen S (2016) Simplified soft sensing model applied in the centralized regenerator of a distributed operating liquid desiccant dehumidification system. *Ind Eng Chem Res* 55:9256–9266. <https://doi.org/10.1021/acs.iecr.6b01988>
- Xiao Y, Liu JJ, Hu Y, Wang Y, Lai KK, Wang S (2014) A neuro-fuzzy combination model based on singular spectrum analysis for air transport demand forecasting. *J Air Transp Manag* 39:1–11. <https://doi.org/10.1016/j.jairtraman.2014.03.004>
- Xin B, Chen J, Zhang J, Fang H, Peng Z-H (2012) Hybridizing differential evolution and particle swarm optimization to design powerful optimizers: a review and taxonomy. *IEEE Trans Syst Man Cybern Part C (Appl Rev)* 42:744–767
- Xu A, Gao F, Wang J, Zhang L (2004) Collision avoidance algorithm based on ANFIS (Beijing Hangkong Hangtian Daxue Xuebao). *J Beijing Univ Aeronaut Astronaut* 30:670–673
- Xu AD, Fan YH, Li ZQ (2011) Modeling of switched reluctance motor based on GA-ANFIS (Dianji yu Kongzhi Xuebao). *Electr Mach Control* 15:54–59
- Yaghoobi A, Bakhshi-Jooybari M, Gorji A, Baseri H (2016) Application of adaptive neuro fuzzy inference system and genetic algorithm for pressure path optimization in sheet hydroforming process. *Int J Adv Manuf Technol* 86:2667–2677. <https://doi.org/10.1007/s00170-016-8349-2>
- Yildiz A, Akin M, Poyraz M, Kirbas G (2009) Application of adaptive neuro-fuzzy inference system for vigilance level estimation by using wavelet-entropy feature extraction. *Expert Syst Appl* 36:7390–7399. <https://doi.org/10.1016/j.eswa.2008.09.003>
- Yin XY (2014) Short-term load forecasting of microgrid by a new advanced prediction strategy, vol 508. <https://doi.org/10.4028/www.scientific.net/AMM.508.231>
- Yousefi M, Mosalanejad M, Moradi G, Abdipour A (2012) Dual band planar hybrid coupler with enhanced bandwidth using particle swarm optimization technique. *IEICE Electron Express* 9:1030–1035
- Zadeh LA (1965) Fuzzy sets. *Inf Control* 8:338–353
- Zahraee S, Assadi MK, Saidur R (2016) Application of artificial intelligence methods for hybrid energy system optimization. *Renew Sustain Energy Rev* 66:617–630
- Zanganeh M, Yeganeh-Bakhtiary A, Bakhtyar R (2011) Combined particle swarm optimization and fuzzy inference system model for estimation of current-induced scour beneath marine pipelines. *J Hydroinf* 13:558–573. <https://doi.org/10.2166/hydro.2010.101>
- Zanganeh M, Mousavi SJ, Etemad-Shahidi A (2006) A genetic algorithm-based fuzzy inference system in prediction of wave parameters. In: Computational intelligence, theory and applications: international conference 9th fuzzy days in Dortmund, Germany, Sept. 18–20, 2006 Proceedings, pp 741–750. https://doi.org/10.1007/3-540-34783-6_72
- Zangeneh AZ, Mansouri M, Teshnehlab M, Sedigh AK (2011) Training ANFIS system with de algorithm. In: Proceedings of 4th international workshop on advanced computational intelligence, IWACI 2011, pp 308–314. <https://doi.org/10.1109/IWACI.2011.6160022>
- Zeinali M, Mazlan SA, Abd Fatah AY, Zamzuri H (2014) A GA-weighted adaptive neuro-fuzzy model to predict the behaviour of magnetorheological damper, vol 663. <https://doi.org/10.4028/www.scientific.net/AMM.663.203>
- Zhang HY (2013) Modeling permanent magnet synchronous motor system in electrical automation engineering based on adaptive neuro-fuzzy inference system, vol 676. <https://doi.org/10.4028/www.scientific.net/AMR.676.297>
- Zhang Y, Wang S, Wu L (2010) A novel method for magnetic resonance brain image classification based on adaptive chaotic PSO. *Prog Electromagn Res* 109:325–343
- Zhang J, Ma W, Ma L (2014) A fault diagnosis method based on ANFIS and bearing fault diagnosis. In: Proceedings of 2014 international conference on information science, electronics and electrical engineering, ISEEE 2014, pp 1274–1278. <https://doi.org/10.1109/InfoSEEE.2014.6947876>
- Zhang Y, Wang S, Ji G (2015) A comprehensive survey on particle swarm optimization algorithm and its applications. *Math Probl Eng* 2015
- Zhang W, Zhu J, Kong LF (2011) Gradient genetic algorithm-based performance fault diagnosis model. In: 2011 2nd international conference on artificial intelligence, management science and electronic commerce, AIMSEC 2011 - Proceedings, pp 3059–3062. <https://doi.org/10.1109/AIMSEC.2011.6010842>
- Zuo L, Hou L, Zhang W, Geng S, Wu W (2010) Application of PSO-adaptive neural-fuzzy inference system (ANFIS) in analog circuit fault diagnosis, vol 6146 LNCS. https://doi.org/10.1007/978-3-642-13498-2_7